

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Yunus Emre Cogurcu, Emrehan Metin

- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.540 to 0.617
- None of 20 transferability diagnostics is shown to order transfer skill

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Yunus Emre Cogurcu^{a,*}, Emrehan Metin^a

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness separates a fire year from a normal year, but models built on it are rarely tested outside the region where they were fitted. Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation, adding six thermal predictors to a terrain, fuel and greenness baseline. The first result is about **evaluation geometry**: holding the model, its predictors and its fitting fixed and changing only which cells are scored, moving from the whole region to the burn scar and its 2 km collar costs **0.143 ROC-AUC** [**+0.077, +0.208**], which exceeds the predictor-block increments this literature reports as findings. A control isolates the cause as the composition of the negative pool rather than class balance: matching prevalence costs -0.000 [$-0.003, +0.002$] while substituting fire-adjacent negatives costs $+0.155$ [$+0.093, +0.217$]. Applied between regions the same effect withdrew three of our own claims. The five study areas enclose very unequal far fields, and restricting each to a 10 km collar, which drops no burned cells, lifts mean transfer from 0.540 to 0.617 and leaves no sign reversal supported under our own criterion, though the

*Corresponding author.

Email address: `ycogurcu@cu.edu.tr` (Yunus Emre Cogurcu)

LST anomaly still differs between regions on a difference interval. The agreed direction is the opposite of the one dryness physics predicts: hotter pre-fire surfaces burned less in every region, and on mutual adjustment temperature survives where greenness does not, so the absolute channels behave here as static land-surface descriptors. Local skill nonetheless does not travel. The thermal block adds +0.056 to +0.153 within regions under blocked cross-validation but +0.022 $[-0.032, +0.077]$ when a whole burn scar is withheld and +0.004 $[-0.028, +0.036]$ across twenty transfer directions, with a sign that varies by pair; the static baseline transfers no better, at 0.537 against 0.541; and equalised transfer of 0.617 sits against a matched within-region reference of 0.772, a shortfall of 0.155. Of twenty candidate diagnostics from five families, none was shown to order transfer: eighteen failed on the frames as drawn, and the two that succeeded require target labels and become degenerate once the frames are equalised. Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [29]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from
5 erature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [42, 14]. The dominant pattern is well established. Geospatial predictors are assembled over a

study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [5]. The resulting
10 susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They
15 are also, as we show, not portable, and a large part of this paper is about how much harder that is to establish than it looks: the reading we first reached, that the relationship is reparameterised locally and points in different directions in different places, does not survive a correction to how the study regions were evaluated (Section 4.10). The dynamic block is worth +0.056 to
20 +0.153 ROC-AUC inside every region under blocked cross-validation, +0.022 when a whole burn scar is withheld, and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own
25 earlier framing took it. The data do not support it. The static baseline transfers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 of transfer on an interval that also spans zero. Two nulls on the portability axis are not an exchange. What is measured is a local
30 gain that does not export, and a local cost when the reversing predictors are removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area
35 and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [33, 31] and is addressed by spatially blocked cross-validation [39, 25]. Blocking is the right correction for what it corrects, but it does not make the estimate honest about a fire
40 the model has not seen, and Section 4.3 measures that gap on identical cells.

Because only that side of the ledger is reported, portability is never entered at all. A predictor block is adopted on the strength of the increment it delivers inside its training footprint, and whether that increment survives a change of region is not asked, even though any regional product built from
45 locally trained models implicitly promises generalisation beyond it. For fire the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not to port cleanly between fire environments [32], and the two recent studies that test model transfer systematically, across Mediterranean countries [8] and US counties [16], both
50 report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific predictor class behaves the same way is the question this paper puts.

1.2. Pre-fire thermal dryness is the natural test case

55 The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors

are static or near-static over the timescale at which fire danger varies, and such a set explains poorly why one summer burned and the preceding one did not. What changes between those summers is the state of the surface, to which satellite thermal observation gives partial access (Section 2.2). The Temperature-Vegetation Dryness Index [35] formalises that access into an internally normalised measure which should, in principle, be less exposed to absolute-temperature offsets between regions than raw land surface temperature. **That theoretical portability advantage is tested here and is not found:** the two internally normalised channels transfer no better than the four absolute ones (Appendix A(f)).

The physics linking moisture stress to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate. That expectation is the motivation for the design, not a finding of it, and Section 4.10 reports that the signed associations run the other way — a hotter pre-fire surface is associated with *less* burning in all five regions, and on mutual adjustment temperature survives where greenness does not, so on this cohort the absolute channels behave as static land-surface descriptors rather than as a dryness index. We keep the framing because it is why these predictors were chosen and why their failure to travel is informative, and we state the contradiction where the evidence appears rather than adjusting the motivation after the fact.

1.3. Why the loss is invisible to the diagnostics in use

A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response

relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [37], an approach with an established remote-sensing literature [38]. Under
85 **concept shift** the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [27].

90 This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [23, 24, 18], are marginal by construction. A target region can sit well inside the training data’s predictor envelope while the relation-
95 ship between those predictors and burning points the other way. This paper measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper, stated here as claims and established with their intervals in Section 4.

100 **Contribution 1. Where a model is scored decides what it appears to know, and the effect is large enough to dissolve findings of our own.** That evaluation extent inflates AUC is established in species distribution modelling (Section 2.5); what is new is a magnitude on the wild-fire problem under a controlled design, and its consequences for a live result.
105 Holding the model, the predictors and the fitting fixed and changing only which cells are scored costs **0.143 ROC-AUC**, which exceeds the predictor-

block increments this literature publishes as findings. A control isolates the cause as the composition of the negative pool rather than class balance (Section 4.3).

110 Applied between regions, the same effect withdraws three claims we had made (Section 4.10). The five study areas enclose very unequal far fields; equalising them to a 10 km collar, which drops no burned cells, lifts mean transfer, reduces the below-chance directions, and leaves no sign reversal supported under this paper’s own criterion — including the two-fire arm
115 inside one study area that we had exempted, wrongly, on the ground that its geography was fixed. The agreed direction is also not the one dryness physics predicts: hotter pre-fire surfaces burned less in every region, and on mutual adjustment temperature survives where greenness does not, so the absolute thermal channels behave here as static land-surface descriptors. The
120 practical consequence is a reporting standard (Section 5.8).

Contribution 2. Local skill does not travel, and correcting the frame does not rescue it. The thermal block is worth a substantial within-region increment in every one of five regions under blocked cross-validation, with every bootstrap interval above zero. Much of that is a property of interleaved holdout: withholding a whole burn scar leaves an increment whose
125 interval spans zero. Across twenty ordered transfer directions its paired contribution also spans zero, with a sign that is a property of the pair rather than of the block, and dropping one region reverses the mean. Two controls bound the reading. The static baseline transfers no better than the dynamic one,
130 so the failure is not the thermal block’s peculiarity; and on matched frames and matched blocking the equalised transfer still falls **0.155** short of the

within-region reference (Section 4.10). The within-region half is not novel, comparable results existing for these landscapes [1, 13]; the paired contrast against portability is. Dimarco et al. [8] transfer a predominantly stationary
 135 predictor set successfully across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

Contribution 3. The shortfall cannot be anticipated by any diagnostic we could run. Twenty candidate diagnostics from five families are evaluated against observed transfer under one bootstrap framework:
 140 predictor-space dissimilarity [23, 24, 18], climatic and geographic distance, learned domain separability, niche-overlap statistics and burn-pattern regime distances. Eighteen were not shown to order the matrix on the frames as drawn; the domain classifier is at ceiling and therefore carries no ordering information at all. Two conditional variants did order it, but both need burned
 145 labels on both sides, both rest on a data-selected feature subset whose all-nine counterparts span zero, and **Contribution 1 removes even those:** recomputed on comparable frames the winning index is unanimous, with no variance left to correlate (Section 4.10). The point survives without any ranking, at the point estimates: the pair with the highest burned-niche overlap
 150 transfers worst in the matrix while the lowest transfers best. That contradicts an expectation the fire literature carries [8, 16] and agrees with species distribution modelling [41, 34].

Two consequences follow, reported as supporting results rather than leading ones. Label-free alignment by standardisation and covariance alignment
 155 [37] does not repair transfer; it compresses most directions towards chance, and we believe this is the first application of covariance alignment to fire sus-

ceptibility. And because the residual is conditional, the resource that closes it is target labels, whose price is measured in Section 4.11.

Alongside these, a leakage-audited, spatially blocked evaluation and transfer protocol is released with code, configuration and frozen outputs, so that
160 most of this result can be re-run rather than taken on trust. One component is an exception and is named in the data-and-code availability statement: the reproduction-check code is not in the released repository at the commit of record. The few-shot export, previously also listed as an exception, was
165 regenerated at commit 6d7a6a71 and its validator now passes on every check. This matters given evidence that wildfire transfer conclusions are sensitive to evaluation design and task formulation [45].

A companion paper treats the observational layer beneath this one, where preprocessing decisions taken before any model is fitted are shown to carry
170 their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study regions, the data, and the modelling, cross-validation, transfer and adaptation protocol. Section 4 reports results. Section 5 discusses them, including limitations. Section 6 concludes.

175 **2. Related work**

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both methodological and application perspectives [42, 14]. The dominant pattern is stable: geospatial predictors are assembled over a study region, paired with a
180 historical burned-area record, and fitted with a supervised classifier, most

often a random forest [5], and the resulting surface is published with a cross-validated AUC in the 0.85 to 0.95 range. That figure is a within-region estimate. Comparable within-region results already exist for the landscapes studied here [1, 13].

185 2.2. *Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through fuel moisture content [47]. Pairing land surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [7], and the Temperature-Vegetation Dryness Index [35] formalises that feature space into an internally normalised measure, which is why it is often expected to travel between regions better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established [19, 21, 20]. Gelabert et al. [9] found dead fine fuel moisture and its anomalies to be the most influential predictor of human-
190 caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. *Spatial validation and transferability*

200 Random cross-validation over spatially autocorrelated cells inflates skill estimates [33, 31], and spatially blocked designs are the standard remedy [39, 25]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [43, 26, 6]. This paper uses spatial

blocking throughout and reports every result at three block sizes, so that a
205 reader can see which verdicts depend on the choice.

Blocking corrects the inflation that random folds produce within a region.
It does not speak to performance on a fire the model has not seen, which
Section 4.3 measures on identical cells. For that, the field’s instrument is
the area of applicability and related predictor-space dissimilarity measures
210 [23, 24, 18], which ask whether a target’s predictor values fall inside the
training data’s envelope. Species distribution modelling has examined trans-
ferability far more systematically than fire has [46], and two studies there
report that geographic and environmental similarity do not reliably predict
transfer success [41, 34], and the transferability of satellite-derived empirical
215 models is an active question in fire remote sensing itself [22].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor
distribution moves but the predictor-response relationship holds, from con-
cept shift, where the relationship itself changes [27]. Covariate shift is in
220 principle correctable without target labels, by per-region standardisation or
by covariance alignment such as CORAL [37], and domain adaptation has
an established remote-sensing literature [38, 30]. Concept shift is not: no
realignment of input distributions can repair a reversal in the sign of an
association, because detecting it requires the labels being withheld. Shift
225 decomposition in applied remote sensing is not itself new [12]. We are not
aware of a prior application of covariance alignment to fire susceptibility, fire
occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [32] re-
230 port, in a Peruvian case study, that meteorologically derived danger indices
do not port cleanly between fire environments. WildfireGenome [16] runs
a leave-one-county-out matrix across seven US counties and reports strong
within-county performance with highly variable off-diagonal transfer, similar
pairs transferring well and dissimilar pairs collapsing; its label, however, is
235 a principal-component composite of hazard *indicators* rather than observed
burned area, and it applies no adaptation. Xu et al. [45] argue that wild-
fire transfer conclusions depend strongly on evaluation design, a caution we
address by fixing the protocol in a project log before the diagnostics were
computed and by reporting every sensitivity axis; that log is not a formal pre-
240 registration, and Section 3.11 states what was fixed and when. Kondylatos
et al. [15] provide Mesogeos, a 1 km Mediterranean datacube for data-driven
wildfire modelling.

Evaluation extent and AUC. Species distribution modelling settled
long ago that the area a model is evaluated over is not a neutral choice. Lobo
245 et al. [17] make it the fifth and, by their own ranking, most important rea-
son to distrust AUC as a comparative measure: the extent of the modelled
area governs how many easy absences enter the calculation, and therefore
the score. VanDerWal et al. [40] show the same lever on the calibration side,
that how far background points are drawn from known presences changes
the fitted model and its apparent accuracy, and Barve et al. [3] give the
250 argument its general form as the accessible area. This literature is qualitative
about magnitude, because magnitude is problem-specific: it establishes that

the effect exists, is directional, and can be large, without fixing a number for any given application. Two things follow for the present paper. First, our
 255 Contribution 1 is a measurement inside that framework rather than a new phenomenon, and Section 1 states it that way. Second, the wildfire literature has largely not imported the lesson: region-wide validation figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells.
 260 The number we obtain, 0.143 ROC-AUC, is larger than the predictor-block increments this literature reports, which is why we treat it as a reporting problem in Section 5.8 rather than a caveat.

The nearest neighbour, and the contrast the present paper draws.
 Dimarco et al. [8] is the closest Mediterranean analogue: 500 m predic-
 265 tors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher. They apply no domain adaptation.¹

Much of the design is shared — Mediterranean regions, 500 m cells,
 270 MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven ignition, ours is burned area, and those are different quantities with different
 275 dominant controls. Read together the two results suggest that the relation-

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

ship between domain similarity and transfer success may be predictor-class dependent, with Vesk et al. [41] and Rousseau and Betts [34] aligning from species distribution modelling while Dimarco et al. and WildfireGenome run against it. We present that as a live disagreement, and Section 5.7 states
280 why our own data do not settle it.

Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology [28]. We follow their Methods, which we take to be authoritative, and note the discrepancy so that the comparison is transparent. Every statement
285 about their work was checked against the full text, read 2026-08-08.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. C.1): Manavgat 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in
290 Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any model metric. One AOI choice was label-informed and is stated as such. The North Evia box was extended after the legacy box was found to carry
295 an atypically high burned prevalence. The extended geometry was then defined from place anchors, and the legacy variant is kept as a sensitivity arm (Appendix A). The AOIs are deliberately not clipped to fire perimeters, so that unburned cells around each fire form the negative class rather than being excluded by construction. A sixth region, Kozan 2023, is carried as a

300 negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths vary between regions with the event: 57 to 61 days for predictors and 35 to 59 days for labels,
 305 counted inclusively. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are reproduced here.

Table 1: Study regions, areas of interest and temporal windows. Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

3.2. Burned-area label and the ~ 500 m analysis grid

310 Labels come from MODIS MCD64A1 Collection 6.1 [10] retrieved through Google Earth Engine [11], whose omission and commission characteristics [4]

bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. The purpose is to separate burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. Predictor variables

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window, through the chain shown in Fig. C.2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL

335 band with the water bit deliberately preserved; the coarse-resolution thermal
input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- 340 • **Land cover** is ESA WorldCover v200 [48], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- **LST anomaly**, a z-score of the current-window LST median against
345 the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index [35] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of
350 twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference**
355 **ence** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than

standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.

- 360 • **Downscaled LST and fused LST.** A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled
365 share is 0.11 % to 9.70 % by region. The downscaler’s own inputs include coordinates, which is the one route by which a coordinate-derived surface re-enters a feature set from which Section 3.14 excludes coordinates. Appendix A(g) reports the increment without these two channels.

370 3.5. *Cell aggregation, validity and analysis populations*

Cell-level values are means over the ~ 510 m cell, computed from valid 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_modeling` when it meets a 30 % valid-pixel floor and no thermal channel is missing. Two populations are then defined. The **primary** population
375 is natural vegetation, cells whose combined tree, shrub and grass fraction reaches 0.50, which excludes cropland from every burnable mask. The **secondary** population is all valid cells, reported throughout as a sensitivity. Every headline result is given for both.

3.6. *Feature sets and classifier*

380 Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the

six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is a random forest [5] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper is confounded by a model choice.

3.7. *Spatial-block cross-validation and bootstrap uncertainty*

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B by col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval’s reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

3.8. *Cross-region transfer protocol*

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal block’s contribution is a paired difference within a direction rather than a

comparison across directions. All twenty ordered directions among the five regions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

410 **Region-wise z-score.** Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

415 **CORAL after region-wise z-score.** Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [37], with regularisation $\lambda = 10^{-5}$ and eigenvalues floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned
420 source. Neither variant sees a target label, and both are verified label-blind at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies
425 outside the released sweep and is reported separately in Appendix A(b).

3.10. Transfer-gap decomposition and the concept-shift diagnostic

For each direction the gap between the target’s own within-region skill and the raw transfer result is split in two. One part is what the best label-free adaptation recovers, and the other is what it does not. The recovered fraction

430 is defined as (adapted – raw) / (within – raw), signed and unclipped, and its interval comes from the same paired bootstrap. The recovered part bounds what covariate-level correction can achieve. The remainder is unrecovered by that correction; Section 4.3 shows it should not be read as a conditional residual, because much of it is incurred inside a single region.

435 The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as weakness. A reversal is called bootstrap-supported only when the two regions’ point estimates fall on
440 opposite sides of 0.5 **and each region’s own interval excludes 0.5**, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the two regions’ intervals to be disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not been shown to point anywhere in that region, so it is recorded as a point reversal only. Appendix B states
445 the rule again beside the counts, and `conditional_similarity_transfer.json` carries it as machine-readable metadata.

3.11. Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are computed for every region pair and rank-correlated (Spearman) against observed raw transfer
450 AUC, under one bootstrap framework that resamples unordered region pairs with both of their ordered directions travelling together. The families are marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity, climatic and geographic distance and a learned domain classifier; burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and con-

455 ditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with the result. Appendix C.2 gives the full specification of each diagnostic and of the bootstrap.

3.12. Interventions

460 **Pooled multi-region training.** Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose
 465 signed association reverses between regions with bootstrap support **on the frames as drawn** are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean** (Bejís against Evia). Section 4.10 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below is reported as a measurement made under
 470 the original protocol rather than as a consequence of an established reversal. They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same
 475 reversal analysis against which the result is then read, so both quantities are post-selection estimates and no correction for that selection is applied.

3.13. Controls on the transfer path

Four evaluations establish what the transfer arms are measuring. All use the transfer protocol of Section 3.8 unchanged — fit on the source cells, apply
480 to the target cells, no refit, no recalibration, no threshold selection — and all were verified to reproduce the frozen cross-region AUCs before being used. They are a **within-region half-split** (the modelled cells cut in two at the median of a grid axis, both axes and both directions, a split discarded when either half is single-class); **leave-one-scar-out** (each burned component of
485 at least 50 cells, dilated by a 2 km buffer, withheld from training and used as the target); a **foreign-region evaluation** of the same held-out scar areas; and the **same blocked model restricted** to those areas, which isolates the evaluation region from the training regime. Buffer variants, the minimum-component rule and the per-split and per-scar positive counts, which are
490 unequal and bear on the interpretation, are in Appendix A(i).

3.14. Leakage control

An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all
495 feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses
500 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region

model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} . All numbers here were produced under scikit-learn 1.9.0 or verified against it; the implementation tolerance that applies if the version is not fixed is stated in Section 5.9(vi), and the companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices, the dependence is reported rather than resolved by choosing the favourable setting.

3.15. *Same-geography event-to-event comparison (Muğla 2021 versus 2022)*

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical analysis grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell spatial-block bootstrap used elsewhere. Season, year and population all differ between the arms, since the 2022 arm is defined by removing the 2021 scar, and Section 4.9 reports what that costs. Appendix C.3 gives the window dates, the population construction and the asymmetry audit.

4. Results

525 4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire candidates and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded from all modelling. The separation is not marginal: the five admitted
530 regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, though one negative control cannot establish it.

535 North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

540 Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region. The increment's bootstrap interval excludes zero in all five regions at 1 km and at 5 km blocking (Fig. C.3). Those are the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154. They rest on 6 to 33 positive-
545 carrying blocks, so they are reported as indicative rather than as intervals. The table note gives the reason.

Table 2: Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Pirri 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

Table 3: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

Table note (resampling units). The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks that carry at least one burned cell. Those counts fall quickly as blocks coarsen. At 550 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No bootstrap replicate was invalid at any block size, with one exception. At 555 20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom of the six positive-carrying blocks just reported, and a further reason to read the row as indicative. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells also feeds only 12 groups into a 5-fold grouped split, so its models train on 560 about ten blocks each. **The 20-cell row of this table should be read as indicative rather than as an interval.** The 10-cell row is the coarsest blocking this design supports properly, because every region there has 16 to 70 positive-carrying blocks. The increment holds at that scale in all five regions. Source: `paper/referee2_numbers.md`, block C, counted from the 565 frozen per-cell prediction tables.

This within-region result is not itself novel. It is reported because the transfer arms below are measured against it.

4.3. What the evaluation frame is worth, within one region

This section isolates the effect that Section 4.10 later applies between 570 regions. It is reported first because it is measured on one model in one region with no transfer involved, so it cannot be explained by anything the

transfer arms do.

Where the skill is lost, on a matched comparison. Four evaluations are reported. The last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first is included to show why an
575 unmatched comparison misleads. Each uses the transfer protocol unchanged: fit, then apply with no refit, no recalibration and no threshold selection.

The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it. What makes that a harder discrimination problem than a whole region is the composition of its negatives, not
580 its burned fraction, and that can be shown rather than argued. Taking the same fitted model and the same out-of-fold predictions, and scoring them on a random sample of region cells drawn at the scar area’s own burned fraction, gives **0.782 against the region-wide 0.782**: matching the prevalence changes nothing, at $-0.000 [-0.003, +0.002]$ over the nine scars. Scoring the
585 same predictions on the scar area itself gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. The whole effect is therefore the negative pool. Every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region’s negatives include its easy far
590 field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is a symptom of that construction, and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions.

All four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded throughout: it is that region’s only component
595 of any size, so holding it out leaves no usable source model and there is no row C for it. Rows A, B and D are reported here on the eight so that the

Table 4: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student t intervals are over the eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one- scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

differences are paired, and the intervals are Student t over those eight, which is the resampling unit for this arm rather than the spatial-block bootstrap used elsewhere in the paper. Eight is a small number and the intervals are
600 wide accordingly. Four of the eight scars are in Muğla and two in Montiferru, so they are not independent; row A in particular is a region-level quantity repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences are computed per scar and paired, which is what they are reported for.

The same model, scored two ways on the same region, differs by
605 **0.143 AUC.** Rows A and B are one model. The only change is which cells it is scored on: the whole region, or the burn scar and its 2 km collar. That change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [+0.157, +0.293] fall from A to C, about two thirds, and it is the size of the entire increment

610 this literature usually reports. Both intervals are Student t over the eight scars, so the "two thirds" is a ratio of two estimates and is compatible with anything from roughly a third to nine tenths; the point of the comparison is the size of the numerator, not the precision of the fraction. A region-wide blocked figure is therefore an upper bound on what the same model achieves
615 where the fire actually is, and the difference is not small enough to ignore. Any comparison of a scar-level result against a region-level reference inherits that, and the region-level reference is what a paper of this kind normally reports.

Neither withholding the fire nor moving 2,800 km has a measurable cost. B minus C, the fire-specific residual on identical cells, is **+0.082**
620 **[−0.011, +0.175]**, which spans zero. C minus D, the effect of replacing a same-region model with one fitted between 306 and 2,802 km away, is **−0.003 [−0.075, +0.069]**, which also spans zero and is an order of magnitude smaller. The second is the cleaner result: on the same cells, a model
625 fitted in another country does as well as one fitted in the same region with the scar withheld, and it survives restriction to Muğla, the only region where holding out one scar still leaves the source model properly trained, at 0.579 against 0.597.

Row D is a mean over four foreign models, and the spread behind it is large. Decomposing it gives **32** scar-by-source combinations
630 running from 0.374 to 0.724, with **ten below chance** and a mean spread of 0.171 across the four sources for a given scar (Appendix A, Table A3). The pooled 0.555 is therefore not a statement that any foreign model does as well as a same-region one, but that the *average* one does, and the variation it

635 averages over is exactly the pair-specific instability that the rest of Section 4 is about.

One confound must be stated. Holding out a region's only large scar also removes most of its positives: the source model retains 11 positives in Evia, 88 in Manavgat and 97 in Montiferru, against about 2,000 in Muğla, 640 which has four separate scars. Those four non-Muğla arms average 0.525 and the four Muğla arms 0.579. Three of the four are genuinely starved — the fourth, Montiferru's second component, retains 472 source positives — so row C mixes "the fire was withheld" with "almost all the positives were withheld". Restricted to Muğla, C is 0.579 and D is 0.597, and the 645 C-to-D comparison still shows nothing. **With eight scars, three of them starved, this design cannot establish a fire-specific residual, only bound it at about 0.18.**

What this establishes is bounded rather than positive. Rows C and D both sit close to chance, so the comparison between them has little dynamic 650 range, and neither the fire's identity nor the region boundary is shown to cost anything here. What is measurable is the change of evaluation geometry between rows A and B. **It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established there.** Nor does it establish the fire event as 655 the unit, because the held-out patch is defined by the labels, so its identity cannot be separated from its location. The within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out's 0.552, with overlapping intervals.

The result is robust to how the held-out patch is defined, at

660 0.543 to 0.565 across minimum component sizes from 25 to 200 cells and under both 4- and 8-connectivity, and flat across buffers of 2, 5 and 10 km.

The increment declines with the holdout, and is not established once the fire is withheld. Contribution 1 is about the paired thermal-minus-baseline difference, so the same evaluations were run on it.

Evaluation	Thermal increment	Interval
Blocked cross-validation, per region	+0.056 to +0.153	every interval above zero
665 Within-region half-split	+0.027	positive in 13 of 18 splits
Leave-one-scar-out	+0.022	[−0.032, +0.077]
Cross-region, twenty directions	+0.004	[−0.028, +0.036]

The point estimate falls monotonically as the holdout hardens, and the last two intervals span zero. Under leave-one-scar-out the increment is positive in six of eight scars and averages +0.022, against a baseline of 0.529 and a thermal 0.552 that are both close to chance on that evaluation; restricted
670 to Muğla it is +0.009 [−0.051, +0.070]. **The within-region increment of +0.056 to +0.153 is therefore substantially a property of interleaved holdout.** It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established there.

675 4.4. Cross-region transfer, and what label-free adaptation does to it

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686 (Fig. C.4). Twelve of 20 directions are above chance with CI support. Six are *below* chance with CI

support: both directions of Manavgat to Bejís and of Manavgat to Muğla,
 680 plus Bejís to Evia and Evia to Bejís. Two intervals span 0.5. Even the best
 raw transfer, Evia to Manavgat at 0.686, stays far below that target’s own
 within-region thermal performance of 0.870. Across all directions the raw
 deficit against the within-region reference is 0.184 to 0.592 AUC. That ref-
 erence is a region-level blocked estimate; the controls below show it is not
 685 matched to a transfer evaluation.

Those counts belong to the 2-cell blocking of Table 5. At the more conser-
 vative 10-cell blocking the same points give 9 above, 4 below and 7 uncertain,
 and no direction changes side of the chance line (Section 4.8). Four of the
 six below-chance directions keep their support there. Bejís to Manavgat and
 690 Manavgat to Muğla lose it and carry no verdict. The qualitative statement
 is unchanged. The counts should not be read as exact.

In precision terms the transfer is worse than the ROC figures suggest. ROC-AUC is the metric used throughout this paper, for compa-
 rability with the literature, but a susceptibility surface is used as a ranked
 695 area budget, so precision-recall is the operational quantity. Across the twenty
 directions the thermal model’s PR-AUC averages **0.156 against a no-skill
 baseline of 0.136**. The mean of the twenty per-direction lifts is 1.16; the
 ratio of the two means just quoted is 1.146. **Six of the twenty fall below
 their own no-skill baseline at the point estimate, and five of those
 700 six have intervals entirely below it.** The exception is Bejís to Manavgat,
 at 0.034 [0.029, 0.042] against a baseline of 0.038, whose interval covers the
 baseline and which is therefore a point-estimate case only. Only one direc-
 tion, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038.

A transferred model therefore ranks burned cells about a sixth better than
705 random on average, and worse than random in five directions with interval
support and a sixth at the point. This is a sharper statement than the ROC
means support and it should be the one a practitioner reads.

The static baseline does not transfer either. This is the control
for the reading the rest of the paper invites, and it constrains that reading
710 sharply. The same twenty directions were run with the terrain, fuel and
greenness baseline alone. The mean target AUC is **0.537**, against **0.541**
for the thermal model. Static attributes of a place are the class Dimarco
et al. transfer successfully, and are the class this paper’s framing treats as
portable. Here that class is itself barely above chance. The paired per-
715 direction contrast is given below and in `baseline_vs_thermal_transfer`
`.csv`. What matters here is that the transfer failure below is not specific to
the dynamic block. A baseline that does not travel, plus pre-fire thermal
state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its in-
720 **terval.** The two matrices were differenced direction by direction. The mean
is **+0.004**, but the mean is not the informative statistic here. The individ-
ual paired contributions span **-0.148 to +0.133**, with **twelve positive**
and eight negative (`baseline_vs_thermal_transfer.csv`). The spread
is thirty times the mean, and the sign is not a property of the block but of
725 the pair it is asked to cross: the same six predictors that add +0.133 in one
direction subtract 0.148 in another. A mean near zero here records cancel-
lation, not consistent absence of effect. The directions are not independent,
because each region appears in eight of the twenty. The interval therefore

depends on what is treated as the resampling unit. All four units the design
730 permits give the same answer:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	$[-0.027, +0.034]$
Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored. Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives
735 $+0.0148, +0.0021, +0.0065, -\mathbf{0.0081}$ and $+0.0060$. **Dropping Evia alone reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

On the frames as drawn, six directions are below chance with interval support, the sharpest at $0.326 [0.305, 0.349]$. No account of merely
740 lost skill produces a reliably reversed ranking, so on this evidence those six would need a mechanism acting on the direction of the relationship, and Sections 4.5 to 4.9 pursue one. **Section 4.10 then shows that most of this count is an artefact of the evaluation frames: equalising them leaves one direction below chance, not six.** The reader should
745 carry that forward through the intervening sections, whose reversal evidence is computed on the frames as drawn. Per-split and per-scar detail is in Appendix A(i).

Label-blind adaptation compresses the matrix toward chance

Table 5: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

rather than repairing it. Under region-wise z-scoring the twenty direc-
750 tions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half
the raw spread, with no adapted direction exceeding 0.631 against the un-
matched within-region references of 0.859 to 0.918. Adaptation raises the
failing directions and degrades most of those that already transferred. Tak-
ing the better of the two adaptations per direction, 14 of the 20 end closer
755 to chance than they began and 6 end further from it; five of those six involve
Montiferru, the smallest and last-added region, and move upward, while the
sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to
6 split should be read at the precision of limitation (ix) in Section 5.9, since
Bejís→Manavgat is counted as compressed on a margin of 0.001.

760 The per-direction decomposition is in Appendix A(j), Table A5; two fig-
ures from it matter here and are used below.

Against the right reference, adaptation is not failing (Fig. C.5).
The three controls above give an achievable reference for a model applied to
a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out.
765 CORAL, committed to in advance, averages **0.552** across the twenty direc-
tions against 0.541 raw, which is exactly that reference. Taking whichever
of the two adaptations scores higher per direction gives 0.556, but that se-
lection uses the target labels the protocol forbids, so it is an oracle upper
bound rather than an achievable result and is reported as one. What it does
770 is regress the matrix towards that value: fourteen of twenty directions move
closer to chance, which harms the directions that already worked and helps
the ones that did not. The verdict that follows is narrower than "alignment
fails". Alignment cannot exceed what a model can achieve on an unseen fire,

and it does not.

775 In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-split control above shows that the larger part of that remainder is already incurred inside the region, so it should not be read as a measure of concept shift. 780 Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

4.5. *Transferability diagnostics: what appears to order transfer, and why it*
785 *does not*

Twenty candidate diagnostics from five families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Geographic separation does not order the matrix on either construction. 790 Over all twenty directions the Spearman correlation between centroid separation and transfer AUC is -0.32 with an interval spanning zero; on the twelve-direction common subset of Table 6 it is -0.24 , also spanning zero. The two nearest directions, Manavgat and Muğla at 306 km, are among the worst on the frames as drawn, while the 2,802 km pair returns 0.326 and 795 0.444, so the transfer mean of 0.541 is not the value at any one separation.

Only two diagnostics have intervals excluding zero, and both are conditional. The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84$ [$+0.58$, $+0.88$]. The cosine variant reaches

Table 6: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) condi- tional	6	+	+0.84 [+0.58 , +0.88] agree- ment fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain- classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] ^{log} effective- N distance	no	no
geographic	1	−	−0.24 [−0.84,	yes	no

+0.81. No marginal measure was shown to order the matrix. That includes
800 area-of-applicability-style dissimilarity in predictor space, climatic distance
and geographic distance. The niche-overlap and regime families were not
shown to order it either. The learned domain classifier is at ceiling, separating
source from target at $AUC \geq 0.96$ for every pair. It always succeeds, which
is why it carries no ordering information.

805 Two limits are stated with the result rather than after it. The first is
size. The index’s tie structure caps the achievable Spearman at +0.861,
so the observed +0.840 sits essentially on that ceiling. Its exact one-sided
permutation p is 0.0060, the smallest this tie structure can produce, against
a Bonferroni threshold of 0.0026 over the nineteen computed variants. No
810 outcome of this diagnostic could have cleared family-wise correction on ten
effective pairs. The second limit is labels. Signed associations need burned
labels in both regions, so the family that appears to work is not available
before deployment, while the family that fails is.

A fourth limit removes the result entirely, and is reported in
815 **Section 4.10.** The sign-agreement index is built from the signed associations
that Section 4.10 shows to be artefacts of the evaluation frames, and it was
correlated against transfer measured on those same frames. Recomputed on
an equalised frame it is unanimous, taking the value 1.0 in every direction
with no variance left to correlate, while the continuous cosine variant falls
820 from $\rho = +0.50$ to +0.12. Everything in this section is therefore reported as
what the original protocol yields, and the conclusion that survives is that **no**
diagnostic tested here was shown to order transfer once the frames
are comparable.

A third limit is about selection, and it is the sharpest of the three. The
825 two rows that clear zero are the *supported-feature* variants. Their feature
subset is chosen by whether two regions’ bootstrap intervals happen to be
disjoint. That is a data-dependent selection, made on the same data, with
no correction. The unselected counterparts over all nine features are $\rho =$
 $+0.50$ $[-0.17, +0.83]$ for the cosine and $\rho = +0.18$ $[-0.40, +0.72]$ for the
830 agreement count. Both span zero. The result lives in the selection step, and
is reported as such.

The families are compared on unequal samples, and equalis-
ing them does not change the ordering. The marginal, applicability,
climatic and geographic rows sit on twelve directions, because those diag-
835 nostics exist only for the four-region subset. The supported-conditional rows
sit on sixteen and the rest on twenty. A reader may reasonably ask whether
the marginal family’s failure is a statement about power rather than about
diagnostics. Every row was therefore recomputed on the common twelve di-
rections. The published values reproduce to 4.8×10^{-5} . The conditional
840 rows still lead, at $+0.87$ $[+0.65, +0.88]$ and $+0.85$ $[+0.43, +0.88]$, and every
marginal row still spans zero. The ordering is not an artefact of unequal
samples.

4.6. *The contrast pair: similarity is not sufficient*

The clearest single view of Table 6 needs no ranking at all (Fig. C.8).
845 Manavgat and Muğla lie in the same country and the same fire year. They
are 306 km apart by the centroid geodesic distance this paper uses as a
diagnostic, and their nearest boundaries are 191 km apart. Their burned cells
occupy the most similar environmental envelope of any pair in the matrix,

with per-feature Schoener’s D of 0.77 to 0.89. Yet on the frames as drawn
850 five of nine feature-response directions point opposite ways. Elevation is one
of them, at a signed AUC of 0.374 [0.289, 0.471] in Manavgat against 0.611
[0.532, 0.690] in Muğla, with disjoint intervals, and all four absolute thermal
channels are among the others. **Section 4.10 withdraws that reading:**
this is the pair whose Manavgat arm is 60 % far field, and under a 10 km collar
855 the elevation figure moves to 0.561 against 0.606, on the same side of 0.5.
The contrast that survives is the transfer result itself, not the reversal count
that was offered to explain it. Transfer is below chance in both directions at
the point estimate on the frames as drawn, at 0.470 and 0.401, but **that half**
does not survive equalisation either: under the 10 km collar the pair
860 transfers at 0.551 and 0.510, both above chance (`aoi_frame_transfer.csv`).
Both directions also sit inside the nominal area of applicability, at 0.875 and
0.531 of target cells, which is itself a full-frame quantity. **What survives is**
the ordinal contrast, and it survives cleanly: on the equalised frame the
most environmentally similar pair still transfers worst in the matrix, at 0.551
865 and 0.510, while the least similar pair reaches 0.669 and 0.624. Schoener’s
D is computed over burned cells only and is therefore collar-invariant, so the
similarity ordering is unchanged. The claim this section supports is that high
envelope overlap does not buy transfer, not that it produces anti-prediction.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes
870 barely overlap, and they carry the most dissimilar values on every overlap
measure. They transfer above chance in both directions. That half of the
contrast rests on point estimates: neither direction carries a verdict at 5
km blocking, and the marginal applicability audit was never produced for

Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee
875 transfer, and dissimilarity does not preclude it. The claim rests on two
coexisting counterexamples, so it does not depend on the number of pairs
available.

4.7. Interventions: pooling and feature removal

(a) **Pooled multi-region training** (Fig. C.6). At the point estimate,
880 training on the pooled primary populations of the other four regions never
beats the best single-source transfer for any target, with shortfalls of 0.02 to
0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two
targets it falls below the pairwise mean and below chance, though only Bejís
is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation
885 does not recover what single-source transfer loses.

(b) **Removing the direction-reversing features.** The two predictors
whose signed association reverses between regions **with bootstrap support
on the frames as drawn** are **elevation_mean** and **lst_anomaly_mean**.
Section 4.10 withdraws that support under an equalised frame, so this selec-
890 tion rule is frame-dependent and the arm below measures what the removal
costs under the original protocol, not a consequence of an established re-
versal. The per-region signed associations and the three pair-level reversals
that meet that criterion are in Appendix B, Tables B2 and B3. Retraining
without them costs **-0.081** of mean within-region AUC, supported in every
895 region (per-region deltas -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every in-
terval entirely below zero), and changes mean transfer by **+0.014** [**-0.017**,
+0.045], which spans zero.

Two qualifications belong with those numbers rather than after them.

First, the debit is not the thermal block's. Dropping elevation alone accounts
 900 for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone
 accounts for -0.013 , from 0.888 to 0.875. Roughly three quarters of the
 cost is therefore the removal of a **baseline** terrain variable, not of a thermal
 channel. Second, both figures are post-selection. The two predictors were
 chosen because they reverse, using the same data on which the -0.081 and
 905 the $+0.014$ are then estimated. No correction for that selection is applied.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

Fig. C.7 shows the four configurations together. What this measures is
 a local cost with no compensating transfer gain. It does not measure an
 exchange, because the transfer side is a null on both arms. The thermal
 910 block's own contribution to transfer is $+0.004$, with an interval spanning
 zero (Section 4.4). Removing the reversing predictors returns $+0.014$, with
 an interval spanning zero. Two nulls on the portability axis are not a price
 paid.

4.8. Sensitivity analyses

915 Eight design choices were varied with everything else held fixed. They
 are the Evia AOI and its prevalence, the CORAL regularisation constant,
 the blocking scale, the closure date of the predictor window, the quality
 screening of the coarse thermal input, the contrast between the normalised

and the absolute dryness channels, the removal of the coordinate-informed
920 channels, and the capacity of the classifier. None changes a conclusion above.
Two bound how the results should be read, so they are carried into the main
text here.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-**
minus-baseline delta verdicts from ten positive, seven negative and three
925 uncertain to six, four and ten. The above-chance verdicts on the thermal
arm itself, a different quantity, move from twelve, six and two to nine, four
and seven. Support is removed from seven verdicts and added to none. The
point estimates are unchanged, but that is an identity rather than a result.
The blocking scale is the resampling unit, and it cannot move an estimate
930 computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-
screened MODIS input. That changes the downscaled surface on 22,304 of
24,150 cells, by up to 10.9 °C. No signed univariate association moves by
more than +0.0003. This closes the one processing-artefact candidate for
935 that region’s behaviour. Appendix A reports all eight arms, including one
that tests a claim of Section 1.2 and does not uphold it.

4.9. The same geography, a second fire, and why it does not survive either

This arm was designed to hold place fixed and vary only the fire, which
would have separated regional concept shift from everything that differs be-
940 tween study areas. Muğla burned twice, in 2021 and again eleven months
later, on the same grid and through the same processing chain, and the
2022 arm is the 2021 population with the 2021 scar removed: 73,098 rows /
2,911 burned for 2021 against 38,790 rows / 331 burned for 2022. **Positive-**

carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm,
945 and eleven is below the sixteen this design sets as its own floor, so the 2022
intervals are read as indicative exactly as the 20-cell row of Table 3 is. Full
per-feature values are in Appendix B, Table B5.

On the frame as drawn, elevation reverses with bootstrap support. In 2021 higher ground burned preferentially, at 0.611 [0.532, 0.690];
950 in 2022 lower ground did, at 0.296 [0.230, 0.355]. The intervals are disjoint
and the difference is -0.317 [-0.414 , -0.220]: the same predictor, region
and grid, and an association pointing the opposite way in two fires eleven
months apart. The four absolute thermal channels change side as well.

Section 4.10 withdraws it. The 2022 arm is one compact scar inside
955 the whole Muğla box, so 93.2 % of its cells lie beyond 10 km of any burned
cell against 55.3 % for the 2021 arm — the most extreme far field in the
cohort. Under the same 10 km collar the 2021 figure barely moves, 0.611 to
0.606 and still supported, while the 2022 figure moves from 0.297 to 0.565,
onto the same side of 0.5 as 2021 with an interval covering chance. The arm
960 shows the same artefact as the cross-region reversals rather than confirming
them. We keep the section because the structural properties below have no
analogue elsewhere in the paper, and because this arm is what motivated the
frame test.

The two arms are not disjoint samples: they share 38,789 of the 2022
965 arm’s 38,790 cells, and elevation, slope and land cover are identical to the
digit across all 73,098 grid cells, so only NDVI and the six thermal channels
carry new information between them. And the removal is the target’s own
positive class, so in the 2022-to-2021 direction not one target positive is in

the source training population while 38,789 of 38,819 target negatives are,
970 and membership of the source training set alone separates the 2021 target's
classes at ROC-AUC 0.9996. **We therefore cannot bound what that
asymmetry does to a transfer estimate between the two arms.** The
known part of the bias runs the safe way: the removed cells are high, with
a median of 563 m against the population's lower centre, so removing them
975 strips high negatives from the 2022 arm and makes the observed reversal
smaller rather than larger.

4.10. The evaluation frame, applied to our own matrix

Section 4.3's effect applies with equal force between regions, and applying
it changes what Sections 4.4 and 4.6 can claim. It withdraws nothing in Sec-
980 tion 4.3 itself, which is what the test is built from. Source: `aoi_frame_auc.`
`csv`, `aoi_frame_transfer.csv`, `collar_frame_bootstrap.csv`, `diagnostics_`
`collar_frame.csv`, and the code deposited under `paper/code/`.

The five areas of interest are not comparable frames. Each region
is a rectangle drawn around a fire, and the rectangles differ by an order of
985 magnitude in how much unburnt far field they enclose.

Montiferru's frame is fire-scale; Manavgat's and Bejís's are roughly three-
fifths far field. The far field is not a neutral addition. In Manavgat the median
elevation of modelled cells rises from 472 m within 5 km of the fire to 955 m
at 10 to 20 km and 1,273 m at 20 to 50 km, against 512 m for the burned cells
990 themselves, in Taurus terrain that no plausible spread model would place at
risk.

**Under an equalised frame the sign reversals of Section 4.6 do
not survive.** Restricting every region to cells within 10 km of any burned

Table 7: Evaluation-frame geometry of the five study regions. Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.csv`; recomputable by `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

cell drops **no positives**, only far-field negatives.

995 On the equalised frame all five regions agree in sign on elevation, on LST
and on TVDI, and both bootstrap-supported elevation reversals of Table
B3 disappear. Applying this paper’s own criterion from Section 3.10, which
requires each region’s own 10-cell block-bootstrap interval to exclude 0.5, **no**
reversal remains bootstrap-supported in the collar frame (`collar_`
1000 `frame_bootstrap.csv`). Two features straddle 0.5 at the point estimate
and neither is supported. `lst_anomaly_mean` runs 0.392 [0.324, 0.460] in
Bejís against 0.584 [**0.497**, 0.669] in Evia, so the point estimates fall on
opposite sides but Evia’s interval includes 0.5 by 0.003. `tvdi_difference_`
`mean` straddles as well, at 0.509 [0.417, 0.597] in Muğla against 0.384 [0.288,
1005 0.504] in Montiferru, and no region’s interval excludes 0.5. These are the two
internally differenced channels, and an earlier draft of this section treated
the first as uniquely informative because it carries no lapse-rate signal; that

Table 8: Signed univariate AUC, frame as drawn against a 10 km collar. Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

argument does not hold, because the second is equally decorrelated from elevation and behaves the same way. Both are point reversals under the
1010 strict criterion.

A weaker instrument does support the anomaly result, and we had given it away. Table B3’s own note commits this paper to a difference interval on the pair as the sharper test, and Section 4.10 had not applied it. Bootstrapping the two regions independently under the collar
1015 and differencing (`matched_frame_gap.csv`), four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean`:

Pair	AUC A	AUC B	Difference	95 % CI
Bejís vs Evia	0.392	0.584	−0.191	[−0.295, −0.083]
Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
Manavgat vs Evia	0.462	0.584	−0.122	[−0.225, −0.020]
Bejís vs Muğla	0.392	0.507	−0.115	[−0.219, −0.002]

Three of the four involve Evia, which is a coherent pattern rather than
1020 scatter, and the anomaly is the one channel the frame-and-terrain mechanism of this section cannot explain, being decorrelated from elevation at −0.195 to +0.256. Three caveats keep it weak and all three are stated rather than buried: nine of ninety feature-by-pair differences clear zero, against about 4.5 expected at nominal 5 % under the null and with no multiplicity correction;
1025 the nine features are effectively two to three dimensions, as the collinearity paragraph below shows; and Evia’s own interval-support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs**

between regions on the weaker difference instrument. We state it that

1030 way rather than quoting the point estimates alone, because the alternative would be to apply a looser standard to the arm that supersedes Table B3 than to Table B3 itself. Elevation under the collar is above 0.5 in all five regions but individually supported in only two, Muğla at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550, 0.740].

1035 **The sign the five regions agree on is not the one the dryness framing predicts.** For LST the common direction is *below* 0.5 in every region, at 0.386, 0.405, 0.332, 0.286 and 0.376: a hotter pre-fire surface is associated with **less** burning, not more, and the same holds for TVDI. This is not a lapse-rate artefact. Pooling concordance within elevation deciles leaves
1040 LST at 0.402, 0.509, 0.363, 0.327 and 0.392, still below 0.5 in four of five regions, and within NDVI deciles at 0.440, 0.486, 0.404, 0.279 and 0.368, below 0.5 in all five, despite $r(\text{LST}, \text{NDVI})$ reaching -0.92 . The earlier version of this section read it as fuel availability dominating surface dryness, on the ground that NDVI is the most consistently supported single predictor
1045 under the collar (0.582 to 0.663, supported in four of five regions). **The reciprocal test does not support that reading and we withdraw it.** Holding LST within deciles, NDVI's own signed association falls to 0.542, 0.574, 0.547, **0.380** and **0.405**, reversing in Evia and Montiferru; holding NDVI, LST never reverses in any region. On mutual adjustment the surviving
1050 channel is LST, not NDVI, so greenness cannot be the mechanism through which the thermal sign acts.

Two further results bound what the sign is. In two of five regions it is a residual spatial gradient: stratifying within deciles of distance to the near-

est burned cell *inside* the collar moves Manavgat from 0.386 to 0.505 and
1055 Montiferru from 0.376 to 0.485, while Bejís, Muğla and Evia hold at 0.350,
0.367 and 0.319. And interval support is not uniform — LST is supported
in four of five regions and TVDI in only three, so "all five agree" is a state-
ment about point estimates. The reading we can defend is that the absolute
thermal channels behave here as **static land-surface descriptors**, corre-
1060 lated within the collar with greenness (-0.55 to -0.92), elevation (-0.46 to
 -0.68) and slope, rather than as a dryness index, and that the two internally
differenced channels, the ones constructed to isolate the dynamic anomaly,
carry no consistent cross-region direction at all. Compositing depth was not
tested and remains an open alternative. Section 1.2 motivates the thermal
1065 block from moisture-stress physics; that motivation is not what the signed
associations show, and Section 5.2 states the consequence.

The same test destroys the one diagnostic that appeared to work. Section 4.5's only diagnostic with an interval excluding zero is the
fraction of features whose signed association points the same way in source
and target, counted over features interval-supported in both. That statistic is
1070 built out of exactly the signed AUCs this section has just shown to be frame
artefacts, and it was correlated against transfer measured on the same un-
equal frames. Recomputing both sides under the 10 km collar (`diagnostics_`
`collar_frame.csv`, `paper/code/verify_diag_collar.py`):

1075 Two scope statements belong with this table. First, **both sides are**
recomputed here under one bootstrap setting, so the comparison is
internally consistent, but the full-frame value is not numerically the pub-
lished one: it is $\rho = +0.86$ over fourteen directions against the $\rho = +0.84$

Table 9: The two diagnostics that ordered transfer, recomputed on an equalised frame. Spearman ρ against target ROC-AUC over the ordered directions in which the diagnostic is defined. Source `diagnostics_collar_frame.csv`; recomputable by `paper/code/verify_diag_collar.py`.

Diagnostic	Full frame	10 km collar
Sign-agreement fraction over supported features	$\rho = +0.86$ ($p = 0.0001$, $n = 14$), values 0, 0.5, 1	1.0 in all 18 defined directions, variance exactly 0 — degenerate
Cosine similarity over all nine signed AUCs	$\rho = +0.50$ ($p = 0.023$, $n = 20$)	$\rho = +0.12$ ($p = 0.61$, $n = 20$)

over sixteen reported in Section 4.5 and Table B1. The support test is itself
1080 bootstrap-dependent, and at this setting three region pairs carry no jointly supported feature rather than two. The argument does not rest on the published figure, it rests on before and after under identical settings. Second, **only these two diagnostics were recomputed on the collar**. The other eighteen already failed to order the matrix on the frames as drawn and
1085 were not rerun; the equalised transfer vector differs substantially from the as-drawn one, so their correlations against it are unknown rather than shown to be null.

Once the frames are equalised every region pair agrees in sign on every jointly supported feature, so the diagnostic has no variance left and its cor-
1090 relation with transfer is undefined rather than weak. This is not a marginal shift: the number of features supported in both regions *rises* from 1.20 to 3.40 per direction, so the diagnostic is better determined and unanimous.

The disagreements it was reading were the far fields.

Contribution 3 is therefore restated, and it becomes a stronger
1095 **negative result.** The finding is not that conditional similarity orders transfer where marginal similarity fails. It is that **no diagnostic tested here has been shown to order transfer** — the eighteen marginal, niche and regime candidates failed on the frames as drawn, and the two conditional variants that did order the matrix stop doing so once the frames are equalised,
1100 because they were measuring how the study rectangles were drawn. The practitioner’s position is worse than Section 4.5 implied, not better: there is no screen, and the apparent exception was an artefact. Section 4.5’s numbers are retained as computed, on the frames as drawn, because they are what a reader following the original protocol would obtain; this section is
1105 the correction.

The two-event arm does not survive the test either, and it is the most extreme case in the cohort. Section 4.9’s arm was exempted on the ground that the study area is fixed; the area is fixed but the evaluation frame is not, and its 2022 arm has **93.2 % of cells beyond 10 km of**
1110 **any burned cell, median 43.6 km** — a larger far field than any cross-region arm. Applying the same collar (`mugla_two_event_collar.csv`; the full-frame values reproduce Table B5):

Muğla arm	share beyond 10 km	full frame	10 km collar
2021	55.3 %	0.611 [0.529, 0.692], supported	0.606 [0.525, 0.685], supported
2022	93.2 %	0.297 [0.229, 0.363], supported	0.565 [0.450, 0.677], not supported

The 2022 arm moves to the **same side of 0.5** as the 2021 arm and its
1115 interval covers chance, while the 2021 arm is nearly frame-invariant, which
isolates the effect to the 2022 arm’s far field rather than to the collar. **No
bootstrap-supported sign reversal survives anywhere in this paper
once evaluation frames are equalised**, between regions or between two
fires in one region.

1120 Two points bound how many independent reversals could have been counted
in the first place. The channels whose reversal vanishes are the ones partly
measuring terrain: current LST correlates with elevation at -0.695 to -0.125
across the regions and TVDI at -0.722 to -0.298 , so their proxy strength
varies fivefold and a frame that shifts the elevation distribution shifts them
1125 with it. And within the collar `fused_lst_mean` correlates with `current_
lst_mean` at 0.99 to 1.00, `downscaled_lst_mean` at 0.97 to 0.99 and `current_
tvdi_mean` at 0.87 to 0.98, so the four absolute channels are close to one
axis and the two differenced channels correlate at 0.64 to 0.94: the "five of
nine directions reverse" count of Section 4.6 counts features, not independent
1130 quantities, and **in effective dimensions it is closer to two**.

The transfer matrix moves as well. Restricting source and target to
the same collar:

Table 10: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use the same 10-cell (≈ 5 km) spatial-block bootstrap on the target as Table 5, 1000 replicates, seed 42. Per-direction bounds are in `aoi_frame_transfer.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / be- low	Paired thermal delta
full	full (Table 5)	0.540	14 of 20	6	9 / 4	+0.003
full	10 km	0.575	17 of 20	3	11 / 1	+0.002
10 km	full	0.571	17 of 20	3	11 / 3	+0.014
10 km	10 km	0.617	19 of 20	1	15 / 1	+0.023
5 km	5 km	0.608	18 of 20	2	12 / 0	+0.014

The reference arm reproduces the frozen matrix, at 0.540 against Table 5's 0.541 and 14 of 20 exactly, so this is measuring the same quantity. The above-
1135 and below-chance counts in this table are point counts. Under the same 10-cell block bootstrap used for Table 5, at 1000 replicates, the full frame gives nine directions above chance and four below with interval support, and the collar frame fifteen above and one below, so the headline movement is six to one at the point estimate and **four to one with interval support** (aoi_
1140 `frame_transfer.csv`, which carries the per-direction bounds). The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

What this does and does not change. Three claims do not survive and are withdrawn here rather than defended. The count of six anti-
1145 predictive directions becomes one. The sign reversal of elevation, LST and TVDI as the mechanism of the residual is a property of the frames, not of the predictor-burning relationship. And the paired thermal contribution is not stable at +0.004: it rises to +0.023 once frames are equalised, so the claim that the block contributes nothing between regions is frame-dependent too.
1150 What survives is the central negative result, but its size must be stated on a matched comparison and an earlier version of this section did not. Quoting the equalised transfer mean of 0.617 against a within-region reference of about 0.87 compares a 10 km-collar number with a full-rectangle one at 1 km blocking, which is the most generous reference in the paper and is the very
1155 figure Section 4.3 argues is an upper bound. Recomputing the within-region reference on the same frame and at the 5 km blocking this design defends (`matched_frame_gap.csv`, `paper/code/verify_matched_gap.py`):

Frame	Within-region (5 km blocking)	Mean transfer	Gap
full rectangle	0.798	0.540	0.258
10 km collar	0.772	0.617	0.155
5 km collar	0.737	0.608	0.129

The shortfall survives on every matched row, which is what matters, but
1160 it is **0.155 at the collar, not the 0.25 the unmatched comparison implies**, and it shrinks as the frame approaches the fire — which is where a susceptibility surface is actually used.

This is Section 4.3’s result turned on our own matrix: we had measured that the composition of the negative pool moves ROC-AUC by more than the
1165 effects this literature reports, then computed a five-region transfer matrix, and a mechanism, on frames whose fire-adjacent share ranges from 37 % to 98 %. Section 5.9(x) records the frame as a limitation of this cohort rather than of the method.

4.11. What target labels cost: the recovery curve

Everything above measures a failure; this prices it. Label-free alignment
1170 does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many, for three regions across all six ordered directions, using one 10-cell (~5 km)
1175 spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the

1180 sixth. That budget is 7 to 20 % of the target’s natural-vegetation population,
which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this
grid’s effective cell area of 0.199 to 0.208 km² (Appendix C) is roughly **540**
to 620 km², so it is a real answer and not a cheap one. Two properties
of the measurement bound it and are stated in the supplement rather than
1185 buried: at the top budget the selection pool is nearly exhausted, so the
narrow upper-budget intervals reflect saturation rather than precision, and
the labelled blocks are drawn from the event being predicted, which is not a
resource available before that event burns. The protocol, the full curve and
the limits are in Supplement S1.

1190 Two limits belong with the number. At small budgets the same interven-
tion damages the direction that already transfers best without any labels.
And six directions in three regions cannot support a general label budget, so
none is offered.

5. Discussion

1195 5.1. *Reading the three findings together*

Section 1.4 states the three findings and Section 4 establishes them; this
section argues from them rather than restating them. One relation between
them does need saying, because it is what makes the paper cohere: the first
finding is not a caveat attached to the other two, it is the instrument that sets
1200 their size. Applied to our own matrix it withdrew three claims we had made,
and what it left standing is a shortfall in transferred skill, not a reversed
relationship.

5.2. *Why the thermal increment is real but local*

The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows that most of the difference between
1205 them is already present inside a single region. The increment should therefore be read as local in a specific sense: it holds where held-out cells are interleaved with training cells, and most of it is gone once they are not, before the fire or the region changes. It is not an artefact to be explained
1210 away — it replicates in five independent regions, survives coarsening of the blocks to about 5 km with its interval intact, persists in the secondary population, and survives a predictor window closed up to two weeks earlier — but it is established under interleaved validation and not established beyond it. The window-closure result is region-specific and should not be overstated:
1215 the contribution strengthens in Bejís and Muğla, is flat in Montiferru, and **weakens monotonically in Evia**, so what holds everywhere is survival, not improvement (Appendix A(d)).

The natural objection is that different Mediterranean regions are simply different systems, so that a predictor meaning one thing in one place and
1220 another elsewhere is a comparison of two systems rather than instability. **We designed the two-Muğla-events arm to answer that objection and it does not answer it.** On the frames as drawn it looked decisive: region, grid, feature registry and processing chain identical, static predictors identical cell by cell, and elevation’s association reversing between the two
1225 fires. Section 4.9 shows the arm is instead the most extreme frame artefact in the cohort — 93.2 % of the 2022 arm’s cells lie beyond 10 km of any burned cell against 55.3 % for 2021 — and under the collar the 2022 figure moves to

the same side of 0.5 as the 2021 arm’s near-unchanged value, with an interval covering chance.

1230 The objection therefore stands unanswered, and three further confounds were never resolved in any case. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells
1235 against a gate minimum of thirty. The population is not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And the 2022 arm rests on eleven positive-carrying 5 km blocks against this design’s own floor of sixteen. With one fire per region everywhere else, and the one arm that held place fixed now withdrawn, **this cohort provides no evidence**
1240 **that the transfer shortfall is regional rather than event-specific**, and Section 5.9(v) is a substantive limit rather than a formality.

One conclusion does survive from the other direction. On the frames as drawn the sharpest supported reversal belonged to elevation, a *baseline* terrain variable, and the static baseline transfers no better than the dynamic
1245 block. Whatever the shortfall is, it is not the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the question that matters when the failure is conditional. A target region can
1250 sit well inside the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability

and transfer the worst in the matrix. On the frames as drawn that pair is below chance in both directions; on the equalised frame it is 0.551 and
1255 0.510, above chance but still the weakest pair in the matrix while the least similar pair reaches 0.669 and 0.624 (Section 4.6). The point does not need anti-prediction: high overlap does not buy transfer.

The same holds for the niche-overlap and regime families. At the point estimates, the pair with the highest burned-niche overlap in the matrix
1260 fails in both directions while the pair with the lowest transfers in both, so no monotone function of overlap can order the outcomes. The qualifier is not decorative: at the 5 km blocking this paper otherwise defends, neither Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its below-chance support, so this contrast is a statement about point estimates
1265 and is made as one. The domain classifier is at ceiling for every pair, which is informative in itself: the regions are trivially separable in predictor space whether or not their transfer works, so separability carries no ordering information here.

5.4. *What the conditional diagnostic is, and what it is not*

1270 The sign-agreement index was the paper’s one diagnostic with an interval excluding zero, and Section 4.5 states its three limits with the result: it sits essentially on its own tie-structure ceiling, it could not have cleared family-wise correction on ten effective pairs whatever it returned, and its feature subset is chosen on the same data, its all-nine-feature counterparts spanning
1275 zero.

A fourth limit removes it altogether. The index is built from signed associations that Section 4.10 shows to be artefacts of the evaluation frames,

and recomputing it on an equalised frame leaves it unanimous and variance-free. So the practical conclusion is not "use this index instead", and it is not
1280 even "the index works but needs labels". It is that **none of the twenty candidates was shown to order transfer**, and the one that appeared to was reading how the study rectangles were drawn. That is a worse position for a practitioner than Section 4.5 alone suggests, and it is the honest one. As always these are nulls on ten effective pairs: not shown to order transfer,
1285 rather than shown incapable of it.

5.5. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below the within-region ceiling, so aggregation does not manufacture the missing
1290 conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and changes mean transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one. The thermal block's own contribution to transfer is $+0.004$, with an interval
1295 spanning zero. Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Both arms are null on the portability axis. What the interventions measure is a local cost and no compensating transfer gain. That is not a conservation law, and it is not a rate at which local skill can be sold for portability. No such rate is estimated here.

1300 5.6. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null. The regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation
1305 can be set aside: cutting Muğla, by far the largest population, to Manavgat's cell count and separately to its positive count leaves Muğla's transfer behaviour inside the subsampling range in both roles. The error was in the hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology [2] in general.

1310 5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [8] transfer successfully across a comparable Mediterranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours describe the state of a surface in one season, which suggests that the relationship between domain similarity and transfer
1315 success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here. Their response variable is human-driven **ignition**; ours is burned **area**. Ignition likelihood is dominated by human access and activity, stable and near-universal drivers, whereas burned extent is dominated by spread. A
1320 predictor-class explanation and a response-variable explanation are therefore confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this cohort the place-attribute class does not travel either.

1325 5.8. *Implications*

For practice, one number should be stated before any other. In precision-recall terms, which is how a susceptibility surface is actually used, transferred models average a PR-AUC of 0.156 against a no-skill baseline of 0.136, and six of twenty directions fall below their own baseline. Whatever the ROC
1330 figures suggest, a model moved to a region it was not fitted in does not usefully rank burned cells there.

This paper also supports one concrete change in what is reported. Alongside a spatially blocked within-region figure, report skill on a held-out burn scar and its surroundings, or on a held-out fire event. On these five regions
1335 the two differ by about 0.14 ROC-AUC on the same model, which is the size of the effect such papers usually claim, so a blocked figure alone should be read as an upper bound. Transfer skill likewise has to be *measured* rather than inferred from similarity, and the pre-deployment diagnostics currently used for that inference did not order it here. Where a model must be moved,
1340 the resource that closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions, and 30 to 57 % in the rest. That is a real answer and not a cheap one, being 7 to 20 % of the target’s natural-vegetation population, and at small budgets the same intervention damages the direction that already transfers
1345 best.

For method development, the results bound what unsupervised alignment can be asked to do. Two label-free methods applied carefully moved fourteen of twenty directions towards chance, and five of the six they moved upward involve Montiferru, the smallest region; the sixth, Manavgat to Muğla, moves

1350 downward. The committed-in-advance CORAL arm reaches a mean of 0.552,
 and taking whichever of the two methods scores better per direction reaches
 0.556 — but that selection uses the target labels the protocol forbids, so it is
 an oracle upper bound rather than an achievable result (Section 4.4). Even
 the oracle only reaches the reference a model can reach on an unseen scar at
 1355 all (Section 4.3), so they are not failing far below an achievable target; they
 are regressing the matrix onto it, which costs the directions that already
 worked. What alignment cannot do is exceed that reference, and a sign
 reversal is not a distribution mismatch that realigning inputs would repair.

5.9. Limitations

1360 The transfer failure is a finding, not a limitation. The limitations are the
 boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say
 how local skill and portability behave for a mixed thermal-plus-weather pre-
 dictor set.

1365 (ii) **The same-geography comparison covers one region only**, and
 even there year and seasonal phase are confounded, a confound that cannot
 be resolved in this study area for the reason given in Section 5.2. Its 331
 burned cells also leave the thermal reversals unresolved at interval level,
 and the pair holds place fixed but not population. Those two arms are
 1370 additionally the only transfer directions here computed by us rather than
 read from the pipeline author’s frozen export, with his unmodified code and
 the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
 [10], whose omission and commission characteristics [4] bound every model

1375 evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072
1380 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported
1385 numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs.** Both the successes and the failures of Section 4.5 should
1390 be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.** It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening of its coarse thermal input, which propagates widely but moves no signed association by more
1395 than +0.0003 (Section 4.8, Appendix A(e)); and the evaluation frame, which explains its elevation figure but not its transfer behaviour (Section 4.10). With one fire season per region the remaining candidates are not separable in this design.

1400 (ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive, seven negative and three uncertain turns on a lower bound of -0.00045 . The point estimates and the sign pattern are stable; the counts are not. Every
 1405 sentence in this paper that leans on an exact count of supported directions should be read at that precision. (x) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it.** Each region is a rectangle drawn around a fire, and the share of modelled cells lying beyond 10 km of any burned cell ranges from 2 % in Montiferru to 63 % in
 1410 Bejís. Section 4.10 shows this is not cosmetic: equalising the frame to a 10 km collar removes both supported elevation reversals, makes all five regions agree in sign on elevation, LST and TVDI, lifts mean transfer from 0.540 to 0.617 and reduces the below-chance directions from six to one. We report the equalised arm alongside the frame-as-drawn arm rather than replacing one
 1415 with the other, because the collar radius is itself a choice and 5 km and 10 km do not agree exactly (0.608 against 0.617). The deeper limitation is that the frames were fixed upstream of this work, in `repo/`, so we can restrict them but not extend them; a region whose rectangle is already fire-scale, Montiferru, cannot be given a far field for symmetry. Any future cohort should fix the
 1420 frame by an explicit accessible-area rule [3] before any predictor is computed, and we treat that as the main design lesson of this paper.

(xi) **One classifier family.** The headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure and least able to extrapolate. Appendix A(h) shows the transfer result is not

1425 an artefact of that choice: three further estimators, including a penalised
linear one, all land between 0.510 and 0.556 and all place fourteen of twenty
directions above chance. Those are point estimates without intervals, so the
ordering among them is not claimed as a result. Other model families were
not tried, and a different inductive bias might behave differently, but within
1430 this family the negative result is a property of the predictors rather than of
an unregularised estimator.

6. Conclusions

Where a fire model is scored decides what it appears to know. Holding the
model, its predictors and its fitting fixed and changing only which cells are
1435 scored, moving from a whole study region to the burn scar and its 2 km collar
costs **0.143 ROC-AUC**. A control isolates the cause as the composition of
the negative pool rather than class balance. The effect is larger than the
predictor-block increments this literature publishes as findings, so a region-
wide validation figure should be read as an upper bound on what a model
1440 achieves where fire actually occurs, and reported as one.

That correction is not only other people's problem. Applied to our own
five-region matrix it withdraws three of our claims. The study areas enclose
very unequal far fields, and equalising them to a 10 km collar, which drops no
burned cells, lifts mean transfer from 0.540 to 0.617, reduces the directions
1445 below chance, and leaves no sign reversal supported under our own criterion
— including the one arm that held place fixed, two fires in the same study
area eleven months apart, which we had wrongly exempted because its ge-
ography was constant while its evaluation frame was not. What had looked

like a reversed predictor-burning relationship was mostly a statement about
1450 how five rectangles were drawn.

What survives is the central result. Pre-fire thermal dryness adds a real
and repeatable within-region increment in all five regions, surviving spatial
blocking at about 5 km and a predictor window closed up to two weeks
before the first labelled burning. It does not travel: on matched frames
1455 and matched blocking, equalised transfer of 0.617 falls 0.155 short of the
within-region reference, the paired cross-region contribution spans zero with
a sign that varies by pair, and the static baseline transfers no better than the
dynamic one, so this is not a peculiarity of thermal predictors. The shortfall
also cannot be anticipated: none of twenty diagnostics from five families was
1460 shown to order the matrix, and the two that appeared to require target labels
and lose their variance once the frames are comparable. Label-free alignment
compresses most directions towards chance rather than repairing them, and
supervised recalibration works but is not cheap.

Two limits should be read with all of this. The agreed thermal direction
1465 is the opposite of the one the dryness framing predicts — hotter pre-fire
surfaces burned less in every region, and holding greenness does not remove
it while holding temperature reverses greenness in two regions — so on this
cohort the absolute channels behave as static land-surface descriptors rather
than as a dryness index. And with one fire per region, and the single arm
1470 that held place fixed now withdrawn, this design measures a transfer shortfall
but cannot attribute it to region rather than to event.

Two practical consequences follow. Report local skill and portability sep-
arately, since a predictor block worth a large within-region increment may

contribute nothing distinguishable from zero across regions. And fix the
1475 evaluation frame by an explicit accessible-area rule before any predictor is
computed, because a frame chosen for convenience can manufacture both
a transfer failure and a mechanism to explain it. Three extensions are left
for future work: temporal transfer, meteorological covariates, and physically
normalised dryness variables.

1480 **Appendix A. Appendix A. Sensitivity analyses**

Each arm below varies one design choice and leaves everything else fixed.
None changes a conclusion in the main text. They are reported so that a
reader can see which choices were tested, and what each was worth. Arm (f)
is the exception in one respect: it is not a robustness check but a test of a
1485 claim the introduction makes, and the claim is not upheld.

(a) Evia AOI and prevalence. The raw transfer arms were repeated
with the legacy, high-prevalence Evia box. Every qualitative conclusion is
unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest
being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the
1490 chance line.

(b) CORAL regularisation, including the canonical $\lambda = 1$. Over
nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves
by at most 0.014 within any direction and 0.008 within the thermal family.
The canonical $\lambda = 1$ of the cited method lies outside that sweep and was
1495 computed separately on the same four directions. It changes the thermal
mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of
−0.005 to +0.016, and none of those four directions crosses the chance line.

The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at $\lambda = 0.1$.

1500 **Those four directions are Bejís–Muğla and Manavgat–Muğla, and they exclude the pair that moves.** An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL result is λ -dependent. A separate frozen artefact, `experiments/cross_`
1505 `region/step10/coral_lambda_sensitivity.csv`, carries it. Bejís to Manavgat gives 0.5571 [0.5292, 0.5857] at $\lambda = 10^{-5}$, 0.5644 at $\lambda = 10^{-3}$ and 0.5473 at $\lambda = 0.1$, all with intervals entirely above 0.5, but **0.4850 [0.4496, 0.5197] at $\lambda = 1$** , which crosses the chance line and loses its interval support. Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

1510 The correct statement is therefore narrower. CORAL-dependent conclusions are insensitive to λ over three orders of magnitude below 0.1; at the cited method’s canonical $\lambda = 1$ the one direction whose adapted interval sits entirely above chance loses that status. $\lambda = 10^{-5}$ is the minimally regularised choice and matches the reference implementation, but the dependence is real
1515 and is reported rather than absorbed.

(c) **Blocking scale.** Recomputing the transfer quantities at 10-cell (≈ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes
1520 support from seven verdicts and adds none. Across five seeds the counts run 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a

result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once
1525 over all target cells, so no choice of block size could have moved one. The comparison does establish two things. Every verdict that changes, changes towards "no verdict". And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this
1530 variation.

(d) Predictor-window closure. The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla,
1535 from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the five regions' MODIS inputs are quality-screened and three are not. The
1540 split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation
1545 is identical in both arms, at 0.374, the population is unchanged, and the within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$. The reason is structural. Elevation is a DEM variable the screening cannot

touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. Details are in `paper/modis_qc_downstream_`
1550 `propagation.md`.

(f) **Normalised against absolute dryness channels.** Section 1.2 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as
1555 a feature-set contrast.

One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Section 4.9, where the absolute channels move. That is a statement
1560 about one region across two fires. **Across regions it does not hold:** `1st_anomaly_mean` is itself one of the two predictors whose reversal is bootstrap-supported on the frames as drawn, reversing between Bejís and Evia (Section 4.10 withdraws that support under an equalised frame), which is why Section 4.7b drops it alongside elevation. Being internally normalised pro-
1565 tects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen ex-
1570 ports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

The advantage is not found. Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.4: its

mean transfer of 0.5371 was recomputed here from the modelling datasets and matches the 0.537 read from the frozen per-direction export. Details are in `paper/anomaly_only_transfer.md`.

1590 The comparison is between two sub-blocks of one thermal set on one cohort. It does not test normalised dryness indices in general, and it does not test a normalisation fitted against a pooled multi-region reference rather than each region’s own baseline years.

1595 **(h) Model capacity.** Every number in this paper comes from one random forest with unlimited depth and `min_samples_leaf = 3`. That is the configuration most able to encode local structure and least able to extrapolate, so a reader may reasonably ask whether the transfer failure is a property of the predictors or of the estimator. Holding the model fixed is right for the internal comparisons and does not license the external claim, so three further estimators were run over the same twenty directions and the same five
1600 within-region folds.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random for- est, depth un- lim- ited, leaf 3	0.888	+0.099	0.541	+0.004	14 of 20
Random for- est, depth 6, leaf 50	0.829	+0.055	0.556	−0.007	14 of 20
Random for- est, leaf 200	0.805	+0.047	0.550	−0.021	14 of 20
Penalised lo- gis- tic re- gres- sion	0.741	+0.045	0.510	−0.024	14 of 20

Regularisation does not rescue transfer. All four estimators land

between 0.510 and 0.556, all four put exactly fourteen of twenty directions
1605 above chance, and the linear model, the one built to extrapolate, transfers
worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741,
which is what regularisation is expected to cost; transfer does not rise to
meet it. The thermal block’s contribution to transfer is +0.004 under the
canonical forest and **negative** under all three regularised alternatives, so the
1610 block does not become portable when the model is made simpler.

The transfer failure is therefore a property of the predictors on this cohort,
not of an unregularised forest. Detail in `paper/model_capacity.json`.

(g) **The coordinate-informed channels.** `downscaled_1st_mean` and
`fused_1st_mean` come from a per-region downscaling model whose own in-
1615 puts include coordinates, which Section 3.14 names as the one route by which
a coordinate-derived surface re-enters a feature set that excludes coordinates.
A coordinate-smoothed surface is by construction locally informative and
non-portable, so if the within-region increment depended on it, that incre-
ment would be this paper’s own result in miniature rather than a finding
1620 about thermal dryness. The arm was therefore run here rather than de-
ferred.

Both channels were dropped and everything refitted, over all twenty direc-
tions and all five within-region folds. The reference configuration reproduces
the frozen exports exactly, at 0.000000 against both the transfer and the
1625 within-region references.

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every region and retains 82 % to 103 % of its size, rising slightly in one region. Mean transfer is likewise unchanged, at 0.545 without the two channels against 0.541 with them and 0.537 for the baseline alone. The local skill this paper reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the companion paper’s figure for the same quantity, 82 % to 103 %, from a separately written harness.

1635 *Appendix A.1. A(i). The four evaluations of Section 4.3, in full*

Section 4.3 reports four evaluations of the same models as a ladder. The per-split and per-scar detail is here, so that the ladder can be checked without leaving the manuscript.

Table A1. Within-region half-split, every split. Source and target positive counts are given because they are unequal, which is the principal limit on this arm: a straight cut does not produce two exchangeable halves. Two splits are unusable because one half of Manavgat contains no burned cells.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table A2. Leave-one-scar-out at a 2 km buffer, every scar. Muğla and Montiferru are the only regions containing more than one burned component of at least 50 cells. Muğla is the only region where holding one out still leaves the source model properly trained across every arm; Montiferru’s two components are very unequal, so one of its arms retains 472 source positives and the other 97. Muğla’s four arms mean 0.579; the four arms in the other regions mean 0.525, and the pooled figure of 0.552 averages the two.

Region	Component	Source	Target positives	Target cells	AUC
		positives left			
evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of the two arms are not comparable with
 each other or with the transfer targets. A held-out scar with its 2 km collar
 contains only fire-adjacent negatives, against a whole target region's inclusion
 of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %,
 are a symptom of that rather than the cause, since ROC-AUC is invariant to
 class balance at fixed class-conditional distributions. The scar arm therefore
 asks for discrimination against the nearest and most similar negatives only,
 while a transfer arm includes the whole easy far field. Section 4.3 states the
 consequence: the last three rows of the ladder are not distinguishable by this
 design.

Table A3. The foreign-region arm, decomposed by source. Each
 held-out scar area is scored with a model fitted on each of the other four
 regions. Table 4's row D is the mean over the **eight** scars that carry a row
 C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this
 appendix is 0.559 against row D's 0.555.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.521	0.631	0.110
North Evia 2021	1	0.590	0.374	0.704	0.329
Manavgat 2021	1	0.468	0.407	0.563	0.156
Montiferru 2021	1	0.504	0.457	0.546	0.089
Montiferru 2021	5	0.490	0.399	0.656	0.257
Muğla 2021	1	0.518	0.425	0.608	0.183
Muğla 2021	6	0.568	0.533	0.597	0.064
Muğla 2021	8	0.632	0.570	0.724	0.155
Muğla 2021	10	0.671	0.579	0.716	0.137

1670 Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724, and ten fall below chance. Restricted to the 32 combinations behind row D the mean is 0.555 and the same ten fall below chance. The mean spread across the four sources for a single scar is 0.164 over the nine scars and 0.171 over the eight. Averaging over sources is what makes row D comparable with
1675 row C, which is fitted on one region; it is not a claim that the choice of foreign source is immaterial, and Section 4.3 states the distinction.

Table A4. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn
1680 at the scar area’s own burned fraction, and on the scar area. Twenty draws per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm and Bejís therefore qualifies.

Target region	Scar	Burned fraction	A whole region	A'	B scar area
				prevalence- matched	
Manavgat 2021	1	0.61	0.797	0.796	0.589
Bejís 2022	1	0.66	0.824	0.825	0.575
Muğla 2021	6	0.72	0.777	0.772	0.564
Muğla 2021	1	0.59	0.777	0.779	0.654
Muğla 2021	10	0.68	0.777	0.779	0.673
Muğla 2021	8	0.57	0.777	0.782	0.761
North Evia 2021	1	0.87	0.864	0.866	0.745
Montiferru 2021	1	0.58	0.720	0.717	0.622
Montiferru 2021	5	0.34	0.720	0.723	0.461
Mean			0.782	0.782	0.627

A minus A', the effect of prevalence alone, is **−0.000** [**−0.003**, **+0.002**].
1685 A' minus B, the effect of replacing the region's negatives with fire-adjacent
ones, is **+0.155** [**+0.093**, **+0.217**]. The evaluation-area effect is entirely
the negative pool.

Appendix A.2. A(j). The transfer-gap decomposition, in full

Section 4.4 states the two figures this table carries — at most 34 % of the
1690 gap recovered in the directions that started below chance, and seven direc-
tions with negative recovery. The table is kept here rather than in the body
because Section 4.3 shows the within-region reference in its denominator is
not matched to a transfer evaluation, and Section 4.10 shows its raw col-
umn is frame-dependent, so the fractions should be read as within-protocol
1695 quantities.

**Table A5. Transfer-gap decomposition (four-AOI set, 12 direc-
tions).** Within = target's within-region thermal AUC; best adapted = the

better of z-score/CORAL; recovered fraction = (adapted - raw)/(within - raw), signed and unclipped, with paired bootstrap CI (1000 replicates).

1700 Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 5. The adapted arms were not recomputed at the coarser blocking of Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
1705 Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.5 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to
1710 check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru.
1715 The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Diagnostic	Family	Exp.	n	dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16		+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16		+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20		+0.50 [−0.17, +0.83]	no
Vector	P(y x) conditional	+	20		+0.27 [−0.36, +0.77]	no
Spearman, all 9	P(y x) conditional	+	20		+0.18 [−0.40, +0.72]	no
Agreement count, all 9	P(y x) conditional	+	20		+0.24 [−0.45, +0.74]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20		+0.22 [−0.42, +0.73]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20		+0.10 [−0.51, +0.68]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20		−0.07 [−0.66, +0.49]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20		−0.23 [−0.75, +0.44]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20		−0.32 [−0.78, +0.33]	no
Domain-classifier AUC	P(x) marginal	−	20		−0.10 [−0.54, +0.43]	no
Predictor-space mean dissimilarity	P(x) marginal	−	12		−0.08 [−0.59, +0.49]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	88	12	+0.22 [−0.48, +0.59]	no
Fraction inside weighted AoA	P(x) marginal	+	12		+0.08 [−0.89, +0.63]	no
Fraction inside unweighted	P(x) marginal	+	12			

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it.

1720 With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

1725 **Reading the two rows that clear zero.** Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and
1730 Section 4.5 says so.

Equal-sample check. The families sit on unequal samples: marginal, applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5}
1735 and leaves the ordering unchanged — the conditional rows still lead at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero. Source: `paper/diagnostics_common_subset.md`.

Appendix B.1. B2. The signed univariate associations the reversal claim rests on

1740 Section 4.7b drops two predictors because their signed association with burning reverses between regions with bootstrap support on the frames as drawn. Section 4.10 shows that both supported elevation reversals disappear

once the frames are equalised, so this table is the evidence for the feature-removal arm and for the narrowed mechanism claim inside Contribution 1,
1745 not for a general reversal mechanism. It was computed for the frozen analysis and is reproduced here because the claim is otherwise asserted rather than shown.

Table B2. Signed univariate AUC of each predictor against burned, by region. Primary natural-vegetation population; 10-cell (~ 5
1750 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_	0.374 [0.289,	0.643 [0.558,	0.611 [0.532,	0.541 [0.448,	0.584 [0.395,
mean	0.471]	0.729]	0.690]	0.626]	0.762]
slope_mean	0.531 [0.423,	0.521 [0.439,	0.637 [0.582,	0.487 [0.418,	0.652 [0.506,
	0.642]	0.605]	0.686]	0.554]	0.771]
ndvi_mean	0.636 [0.587,	0.559 [0.497,	0.662 [0.616,	0.639 [0.575,	0.586 [0.450,
	0.676]	0.619]	0.704]	0.701]	0.704]
lst_anomaly_	0.482 [0.428,	0.418 [0.364,	0.485 [0.395,	0.640 [0.567,	0.395 [0.285,
mean	0.530]	0.480]	0.566]	0.710]	0.535]
current_lst_	0.538 [0.452,	0.477 [0.401,	0.325 [0.271,	0.377 [0.301,	0.370 [0.248,
mean	0.621]	0.547]	0.382]	0.456]	0.513]
current_	0.552 [0.460,	0.517 [0.429,	0.336 [0.275,	0.362 [0.285,	0.356 [0.233,
tvdi_mean	0.641]	0.595]	0.398]	0.442]	0.499]
tvdi_	0.449 [0.391,	0.512 [0.443,	0.490 [0.396,	0.519 [0.444,	0.378 [0.282,
difference_	0.505]	0.583]	0.575]	0.589]	0.497]
mean					
downscaled_	0.552 [0.466,	0.484 [0.400,	0.307 [0.253,	0.377 [0.297,	0.365 [0.240,
lst_mean	0.637]	0.560]	0.366]	0.459]	0.511]
fused_lst_	0.540 [0.454,	0.481 [0.404,	0.325 [0.272,	0.376 [0.300,	0.370 [0.248,
mean	0.622]	0.551]	0.383]	0.456]	0.513]

1755 **What counts as a reversal.** A pair of regions is called a reversal

only when their signed associations point to opposite sides of 0.5 **and each region’s own interval excludes 0.5**. That is stricter than requiring the two regions’ intervals to be disjoint, and the difference matters: for `current_1st_mean` between Manavgat and Muğla the two intervals are disjoint, but
1760 Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no established direction to reverse from. That pair is a point reversal, not a supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not counted. The conservative criterion costs the paper findings rather than
1770 manufacturing them: a difference interval on the pair, which is the instrument Section 4.9 uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked
1775 area budget, so precision-recall is the operational quantity, and at target prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same frozen step9b exports as Table 5.

Table B4. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the target's burned prevalence. Lift is PR-AUC divided
1780 by that baseline; a lift of 1 is no better than random ranking. Ordered by lift.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

Six directions fall below their own no-skill baseline, and only one exceeds twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.10 shows that this count is largely a property of the evaluation frames rather than of a

reversed predictor-burning relationship.

Appendix B.3. B5. The same-geography event pair, in full

Section 4.9 reports this arm and Section 4.10 withdraws its reversal as
1790 a frame artefact. The per-feature values are kept here because the arm is
what motivated the frame test, and because its structural asymmetries have
no analogue in the twenty-direction matrix.

**Table B5. Signed univariate feature-burned AUC, Muğla 2021
versus 2022.** Raw AUC against burned, never folded to $\max(\text{AUC}, 1 -$
1795 $\text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42
(Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and
38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for
the 2021 arm and 11 for the 2022 arm.** Table 3’s note sets sixteen as
the floor this design supports at that blocking, so the 2022 intervals here fall
1800 below the paper’s own standard and are read as indicative, exactly as the 20-
cell row of Table 3 is. The 2022 arm is additionally a single compact scar, so
its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_`
`2021__mugla_2022_event_relative/step9g_direction_reversal_table.`
`csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

Appendix C. Appendix C. Protocol detail

These protocols are given here in full rather than in Methods, because each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) = 17 \times 17$ pixels, giving a nominal cell edge of 510 m, and each cell is identified by its integer block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSC:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as `burn_date_pixel_agreement_fraction` but no agreement threshold is imposed. The

label never affects a cell’s eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region’s own predictor window are removed from its analysis universe
1840 rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

1845 Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead.
1850 The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

1855 Twenty candidate diagnostics from five families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-
1860 response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [23, 24, 18], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [36] and
1865 Warren’s I [44] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are in-
1870 dividually interval-supported.

Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the
1875 effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 1880 versus 2022)

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor
1885 window closing the day before ignition and a 49-day label window opening

on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the previous year’s scar does not enter the following year’s analysis: it removes 3,073 cells, of which 2,941 belong to the primary population, taking it from 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of 38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population. Section 4.9 states what follows for how these two numbers may be read.

CRediT authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work
1910 reported in this paper.

Funding

This work is an output of project 17506, supported by the Scientific Research Projects (BAP) unit of Çukurova University.

Data availability

1915 All satellite inputs are public and are obtained through Google Earth Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods, together with the three components that fall outside that release. No digital
1920 object identifier is minted and no archival deposit exists.

References

- [1] Alkan Akıncı, H., Akıncı, H., 2023. Machine learning based forest fire susceptibility assessment of Manavgat district (Antalya), Turkey. *Earth Science Informatics* 16, 397–414. doi:10.1007/s12145-023-00953-5.
- 1925 [2] Archibald, S., Lehmann, C.E.R., Gómez-Dans, J.L., Bradstock, R.A., 2013. Defining pyromes and global syndromes of fire regimes. *Proceedings of the National Academy of Sciences* 110, 6442–6447. doi:10.1073/pnas.1211466110.

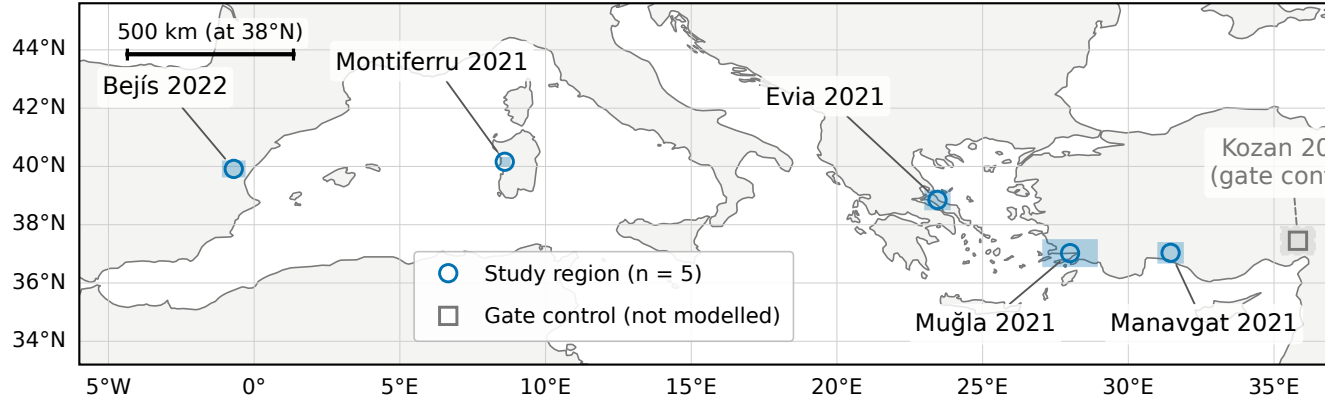


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

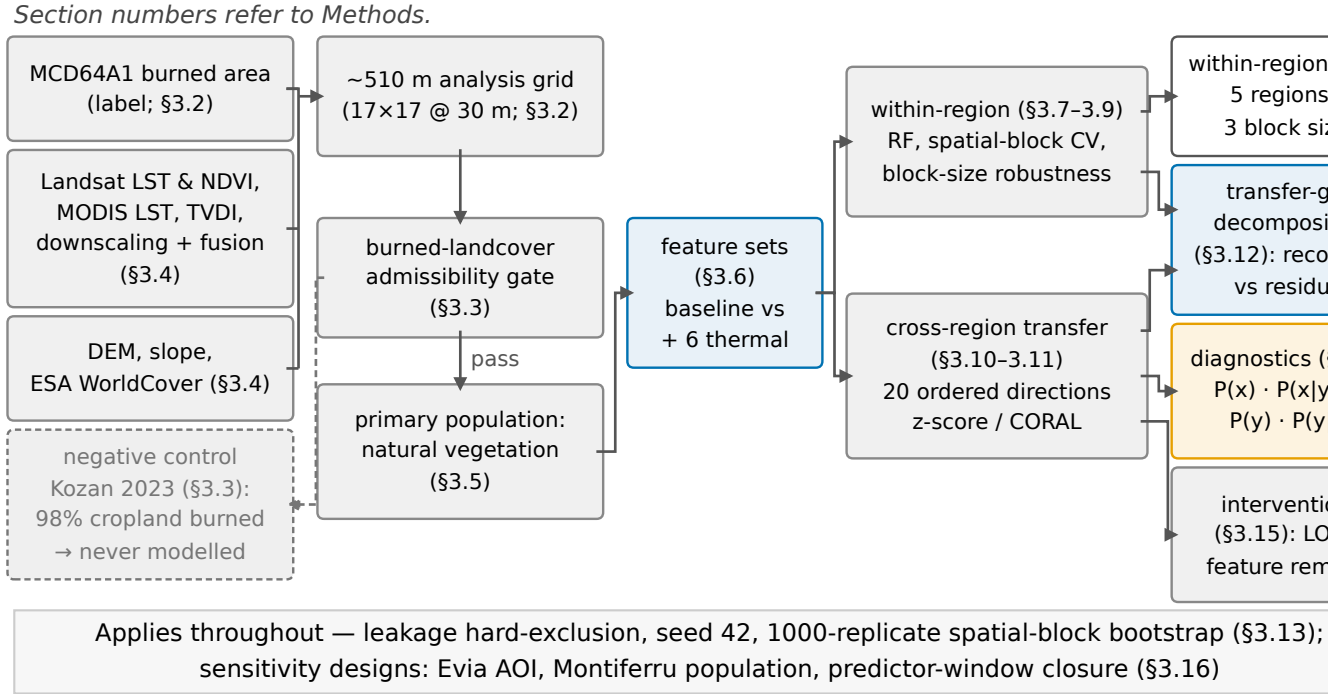


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

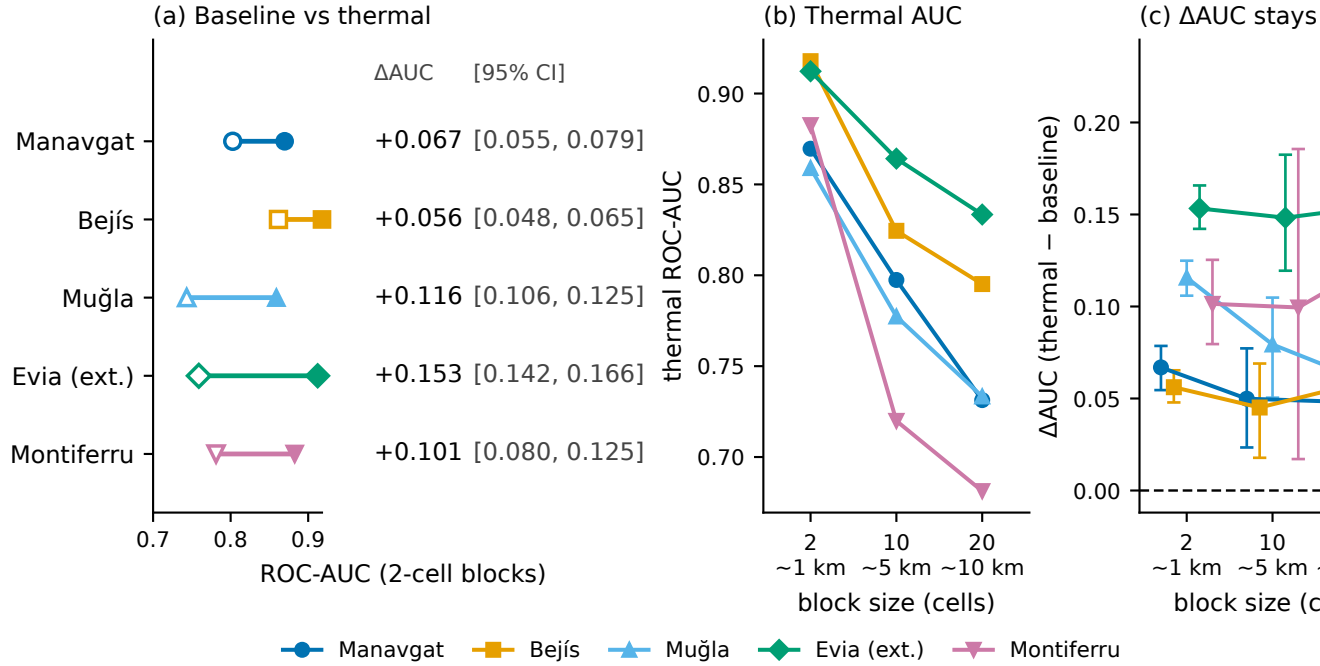


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region–block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 3, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

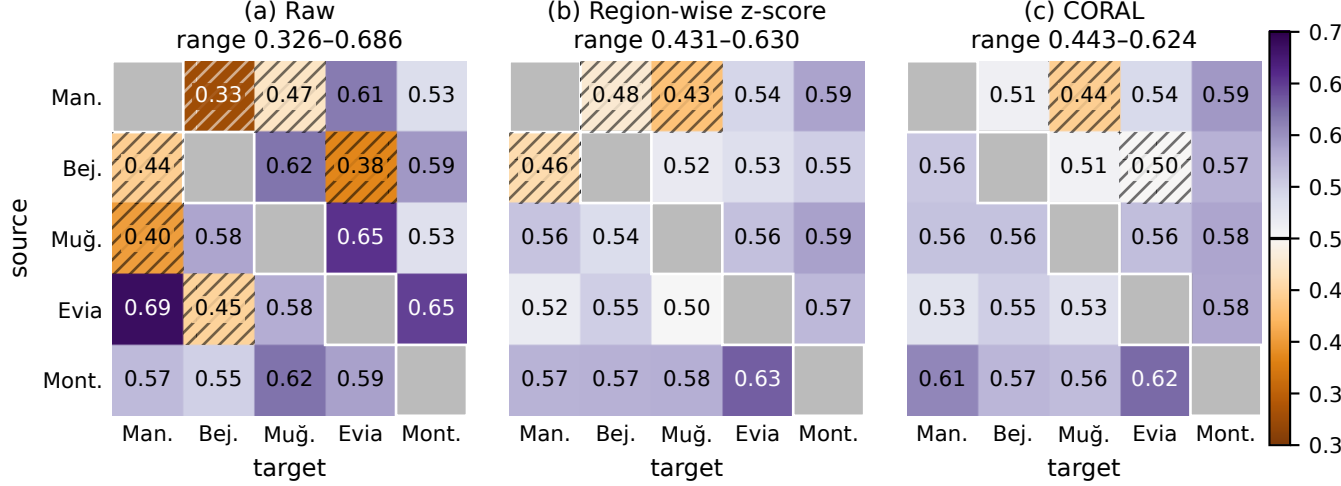


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

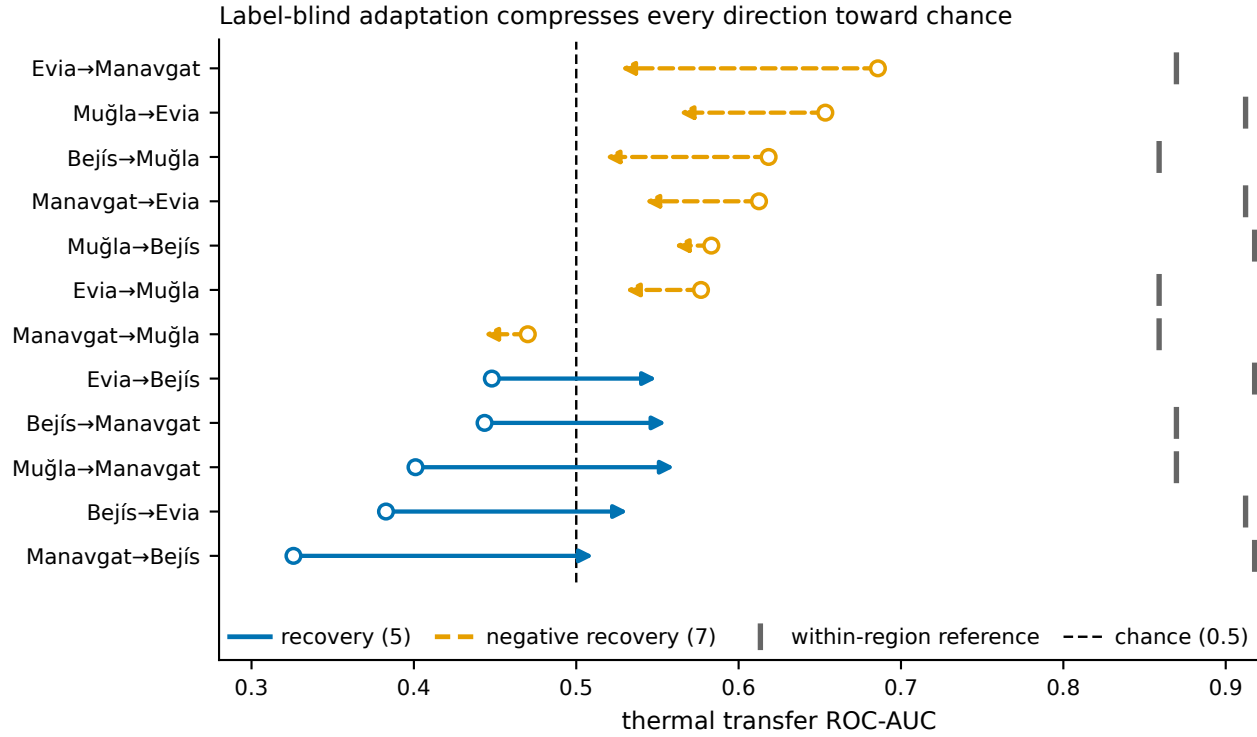


Figure C.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

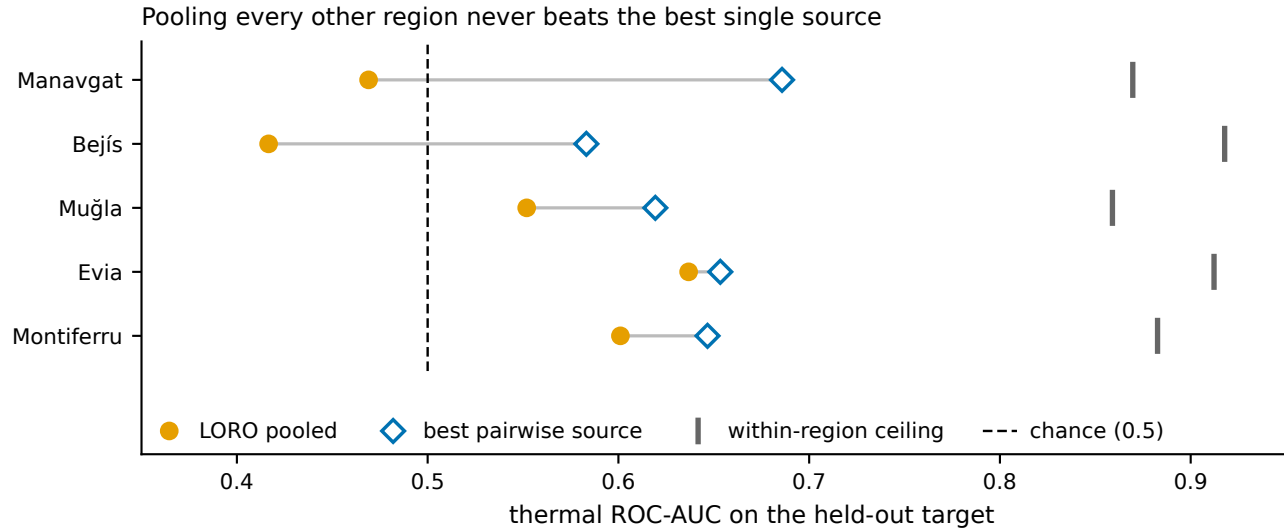


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

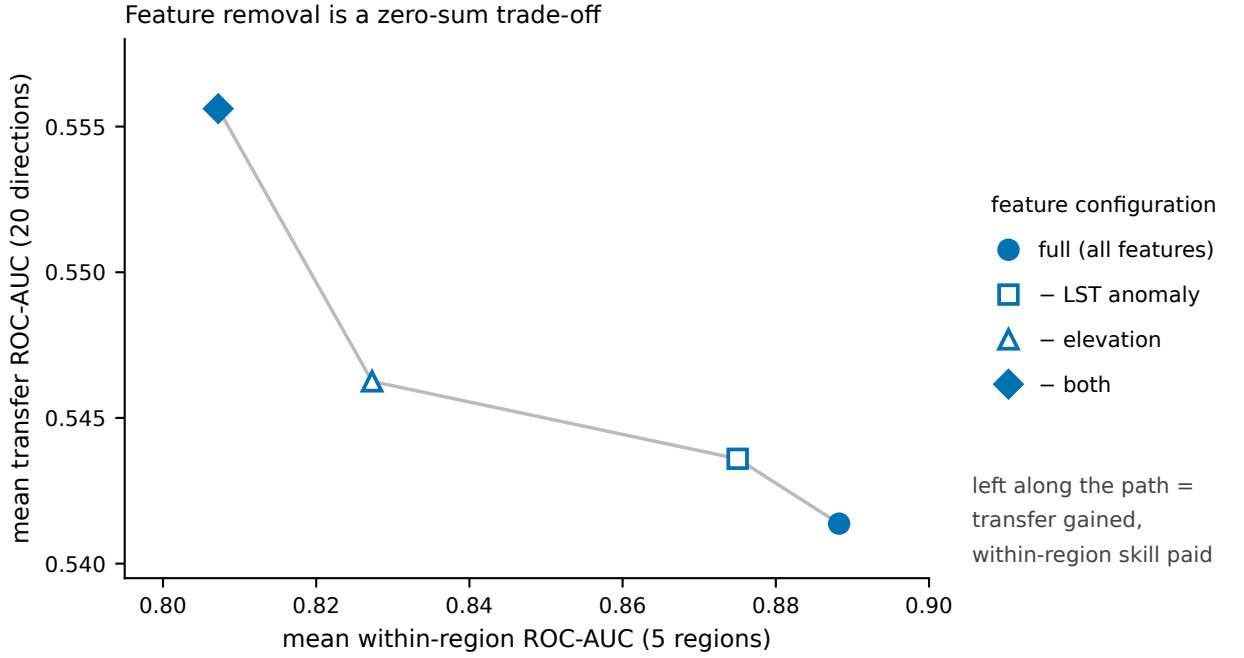


Figure C.7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

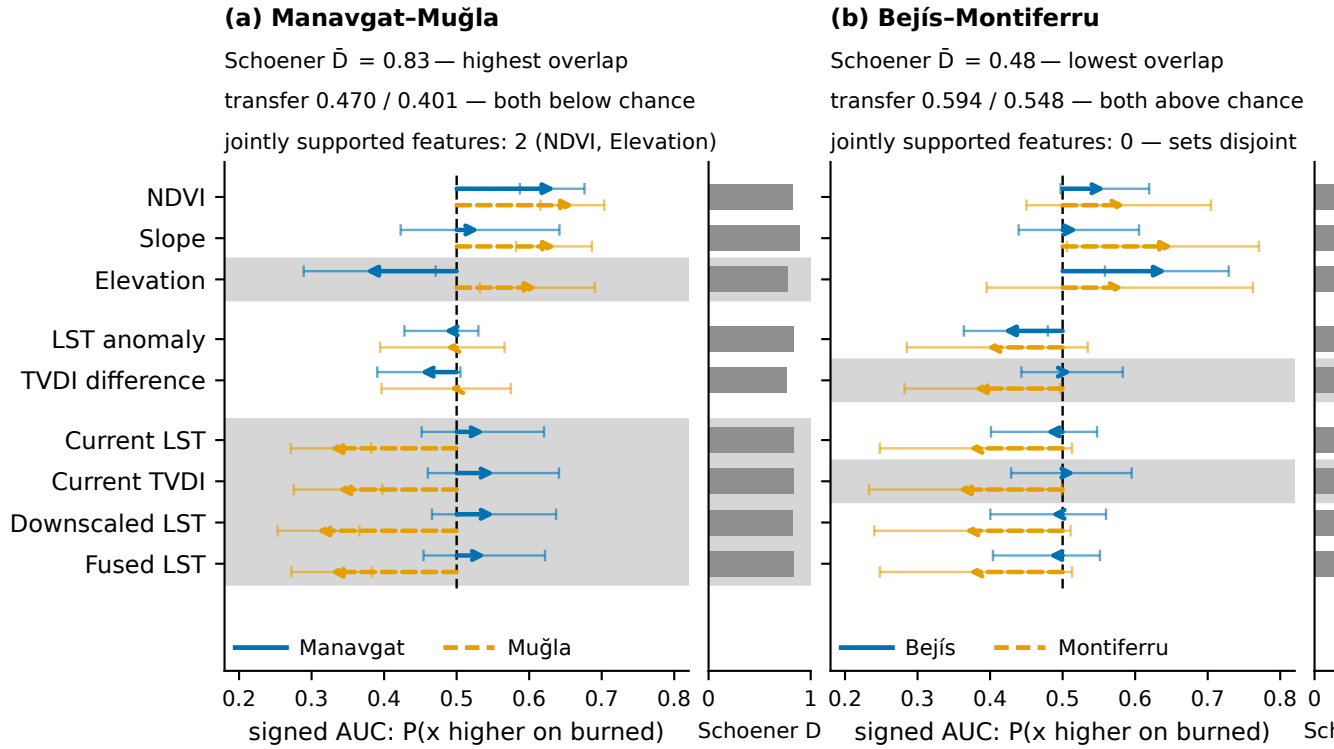


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

- 1930 [3] Barve, N., Barve, V., Jiménez-Valverde, A., Lira-Noriega, A., Mather, S.P., Peterson, A.T., Soberón, J., Villalobos, F., 2011. The crucial role of the accessible area in ecological niche modeling and species distribution modeling. *Ecological Modelling* 222, 1810–1819. doi:10.1016/j.ecolmodel.2011.02.011.
- 1935 [4] Boschetti, L., Roy, D.P., Giglio, L., Huang, H., Zubkova, M., Humber, M.L., 2019. Global validation of the Collection 6 MODIS burned area product. *Remote Sensing of Environment* 235, 111490. doi:10.1016/j.rse.2019.111490.
- [5] Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:1010933404324.
- 1940 [6] de Bruin, S., Brus, D.J., Heuvelink, G.B.M., van Ebbenhorst Tengbergen, T., Wadoux, A.M.J.C., 2022. Dealing with clustered samples for assessing map accuracy by cross-validation. *Ecological Informatics* 69, 101665. doi:10.1016/j.ecoinf.2022.101665.
- 1945 [7] Chuvieco, E., Cocero, D., Riaño, D., Martín, P., Martínez-Vega, J., de la Riva, J., Pérez, F., 2004. Combining NDVI and surface temperature for the estimation of live fuel moisture content in forest fire danger rating. *Remote Sensing of Environment* 92, 322–331. doi:10.1016/j.rse.2004.01.019.
- 1950 [8] Dimarco, N.A., Faraji, I., Wahbi, M., Maatouk, M., Boulaassal, H., Yazidi Aalaoui, O., El Kharki, O., 2026. Generalizing human-driven wildfire ignition models across Mediterranean regions using harmonized

remote-sensing and machine-learning data. *Geomatics* 6, 13. doi:10.3390/geomatics6010013.

- 1955 [9] Gelabert, P.J., Jiménez-Ruano, A., Ochoa, C., Alcasena, F., Sjöström, J., Marrs, C., Ribeiro, L.M., Palaiologou, P., Bentu Martínez, C., Chuvieco, E., Vega-García, C., Rodrigues, M., 2025. Assessing human-caused wildfire ignition likelihood across Europe. *Natural Hazards and Earth System Sciences* 25, 4713–4729. doi:10.5194/nhess-25-4713-2025.
- 1960 [10] Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. *Remote Sensing of Environment* 217, 72–85. doi:10.1016/j.rse.2018.08.005.
- 1965 [11] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. doi:10.1016/j.rse.2017.06.031.
- 1970 [12] Huang, Y., Liu, N., Chen, W., Tan, W., Deng, Y., Wang, Z., Chlus, A., Shen, J., Townsend, P.A., 2026. Enhancing transferability of foliar trait retrieval models: A comparative analysis of transfer learning strategies and domain shift characterization. *Remote Sensing of Environment* 337, 115336. doi:10.1016/j.rse.2026.115336.
- [13] Iban, M.C., Sekertekin, A., 2022. Machine learning based wildfire susceptibility mapping using remotely sensed fire data and GIS: A case

- 1975 study of Adana and Mersin provinces, Turkey. *Ecological Informatics* 69, 101647. doi:10.1016/j.ecoinf.2022.101647.
- [14] Jodhani, K., Bhatiya, L., Sirvi, L., Chanda, A., Thakkar, S., Rathnayake, U., 2026. Artificial intelligence and remote sensing frameworks for wildfire monitoring and risk analysis across multiple ecosystems: A review. *AIMS Environmental Science* 13, 56–98. doi:10.3934/environsci.2026004.
- 1980 [15] Kondylatos, S., Prapas, I., Camps-Valls, G., Papoutsis, I., 2023. Meso-geos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean, in: *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023) Datasets and Benchmarks Track, pp. 50661–50676. doi:10.52202/075280-2204.
- 1985 [16] Liu, C., Mostafavi, A., 2025. WildfireGenome: Interpretable machine learning reveals local drivers of wildfire risk and their cross-county variation. *arXiv preprint arXiv:2511.11589*. doi:10.48550/arXiv.2511.11589, arXiv:2511.11589.
- 1990 [17] Lobo, J.M., Jiménez-Valverde, A., Real, R., 2008. AUC: a misleading measure of the performance of predictive distribution models. *Global Ecology and Biogeography* 17, 145–151. doi:10.1111/j.1466-8238.2007.00358.x.
- 1995 [18] Ludwig, M., Moreno-Martinez, A., Hölzel, N., Pebesma, E., Meyer, H., 2023. Assessing and improving the transferability of current global spa-

- tial prediction models. *Global Ecology and Biogeography* 32, 356–368. doi:10.1111/geb.13635.
- [19] Maffei, C., Alfieri, S.M., Menenti, M., 2018. Relating spatiotemporal
2000 patterns of forest fires burned area and duration to diurnal land surface temperature anomalies. *Remote Sensing* 10, 1777. doi:10.3390/rs10111777.
- [20] Maffei, C., Lindenbergh, R., Menenti, M., 2021. Combining multi-
spectral and thermal remote sensing to predict forest fire characteristics.
2005 *ISPRS Journal of Photogrammetry and Remote Sensing* 181, 400–412. doi:10.1016/j.isprsjprs.2021.09.016.
- [21] Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and
rate of spread from pre-fire multispectral satellite measurements. *ISPRS
Journal of Photogrammetry and Remote Sensing* 158, 263–278. doi:10.
2010 1016/j.isprsjprs.2019.10.013.
- [22] Marino, E., Yáñez, L., Guijarro, M., Madrigal, J., Senra, F., Rodríguez,
S., Tomé, J.L., 2024. Transferability of empirical models derived from
satellite imagery for live fuel moisture content estimation and fire risk
prediction. *Fire* 7, 276. doi:10.3390/fire7080276.
- [23] Meyer, H., Pebesma, E., 2021. Predicting into unknown space? Esti-
2015 mating the area of applicability of spatial prediction models. *Methods in
Ecology and Evolution* 12, 1620–1633. doi:10.1111/2041-210X.13650.
- [24] Meyer, H., Pebesma, E., 2022. Machine learning-based global maps of

- ecological variables and the challenge of assessing them. *Nature Communications* 13, 2208. doi:10.1038/s41467-022-29838-9.
- 2020 [25] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. *Environmental Modelling & Software* 101, 1–9. doi:10.1016/j.envsoft.2017.12.001.
- 2025 [26] Milà, C., Mateu, J., Pebesma, E., Meyer, H., 2022. Nearest neighbour distance matching leave-one-out cross-validation for map validation. *Methods in Ecology and Evolution* 13, 1304–1316. doi:10.1111/2041-210X.13851.
- [27] Moreno-Torres, J.G., Raeder, T., Alaiz-Rodríguez, R., Chawla, N.V., Herrera, F., 2012. A unifying view on dataset shift in classification. *Pattern Recognition* 45, 521–530. doi:10.1016/j.patcog.2011.06.019.
- 2030 [28] Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 4349–4383. doi:10.5194/essd-13-4349-2021.
- 2035 [29] Pausas, J.G., Keeley, J.E., 2021. Wildfires and global change. *Frontiers in Ecology and the Environment* 19, 387–395. doi:10.1002/fee.2359.
- 2040 [30] Persello, C., Bruzzone, L., 2012. Active learning for domain adaptation in the supervised classification of remote sensing images. *IEEE Transac-*

tions on Geoscience and Remote Sensing 50, 4468–4483. doi:10.1109/TGRS.2012.2192740.

- [31] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N.,
2045 Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin,
A., Gourlet-Fleury, S., Pélissier, R., 2020. Spatial validation reveals
poor predictive performance of large-scale ecological mapping models.
Nature Communications 11, 4540. doi:10.1038/s41467-020-18321-y.
- [32] Podschwit, H., Jolly, W., Alvarado, E., Verma, S., Ponce, B., Markos,
2050 A., Aliaga-Nestares, V., Rodriguez-Zimmermann, D., 2022. Reliability
of cross-regional applications of global fire danger models: A Peruvian
case study. Fire Ecology 18, 25. doi:10.1186/s42408-022-00150-7.
- [33] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillerá-
Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller,
2055 W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-
validation strategies for data with temporal, spatial, hierarchical, or
phylogenetic structure. Ecography 40, 913–929. doi:10.1111/ecog.
02881.
- [34] Rousseau, J.S., Betts, M.G., 2022. Factors influencing transferability
2060 in species distribution models. Ecography 2022, e06060. doi:10.1111/
ecog.06060.
- [35] Sandholt, I., Rasmussen, K., Andersen, J., 2002. A simple interpreta-
tion of the surface temperature/vegetation index space for assessment of

- surface moisture status. *Remote Sensing of Environment* 79, 213–224.
 2065 doi:10.1016/S0034-4257(01)00274-7.
- [36] Schoener, T.W., 1968. The Anolis lizards of Bimini: Resource partitioning in a complex fauna. *Ecology* 49, 704–726. doi:10.2307/1935534.
- [37] Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy domain adaptation, in: *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI-16)*, AAAI Press, Phoenix, Arizona, USA.
 2070 pp. 2058–2065. doi:10.1609/aaai.v30i1.10306.
- [38] Tuia, D., Persello, C., Bruzzone, L., 2016. Domain adaptation for the classification of remote sensing data: An overview of recent advances. *IEEE Geoscience and Remote Sensing Magazine* 4, 41–57. doi:10.1109/MGRS.2016.2548504.
 2075
- [39] Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillerá-Arroita, G., 2019. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution* 10, 225–232. doi:10.1111/2041-210X.13107.
 2080
- [40] VanDerWal, J., Shoo, L.P., Graham, C., Williams, S.E., 2009. Selecting pseudo-absence data for presence-only distribution modeling: How far should you stray from what you know? *Ecological Modelling* 220, 589–594. doi:10.1016/j.ecolmodel.2008.11.010.
- 2085 [41] Vesik, P.A., Morris, W.K., Neal, W.C., Mokany, K., Pollock, L.J., 2021.

Transferability of trait-based species distribution models. *Ecography* 44, 134–147. doi:10.1111/ecog.05179.

- [42] Vibhandik, P., Sawant, S., Joshi, A., Bidwe, R., 2026. A systematic literature review on forest fire susceptibility mapping using geo-spatial technology and future research directions. *Discover Artificial Intelligence* 6, 921. doi:10.1007/s44163-026-00921-0.
- [43] Wadoux, A.M.J.C., Heuvelink, G.B.M., de Bruin, S., Brus, D.J., 2021. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling* 457, 109692. doi:10.1016/j.ecolmodel.2021.109692.
- [44] Warren, D.L., Glor, R.E., Turelli, M., 2008. Environmental niche equivalency versus conservatism: Quantitative approaches to niche evolution. *Evolution* 62, 2868–2883. doi:10.1111/j.1558-5646.2008.00482.x.
- [45] Xu, Y., Dai, Y., Chang, L., Wang, Q., Dong, Y., 2026. Does your wildfire prediction model actually work, or just score well? arXiv preprint arXiv:2605.18911. doi:10.48550/arXiv.2605.18911, arXiv:2605.18911.
- [46] Yates, K.L., Bouchet, P.J., Caley, M.J., Mengersen, K., Randin, C.F., Parnell, S., et al., 2018. Outstanding challenges in the transferability of ecological models. *Trends in Ecology & Evolution* 33, 790–802. doi:10.1016/j.tree.2018.08.001.
- [47] Yebra, M., Dennison, P.E., Chuvieco, E., Riaño, D., Zylstra, P., Hunt, E.R., Danson, F.M., Qi, Y., Jurdao, S., 2013. A global review of remote

- sensing of live fuel moisture content for fire danger assessment: Moving
2110 towards operational products. *Remote Sensing of Environment* 136,
455–468. doi:10.1016/j.rse.2013.05.029.
- [48] Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W.,
Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz,
S., Lesiv, M., Herold, M., Tsendbazar, N.E., Xu, P., Ramoino, F., Arino,
2115 O., 2022. ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/
zenodo.7254221.